

Mixed-model association under population structure: numerical foundations, calibration diagnostics and research directions for mixmogam

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Abstract

Mixed-model association combines a model of genetic background with a numerical approximation and a reference distribution for testing. These three parts require separate validation. We describe the revised mixmogam package and examine how population structure exposes their differences. The package provides dense and streaming Gaussian mixed-model analyses, leave-one-chromosome-out (LOCO) scans, and experimental two-step approximations. We derive the dependence of the association denominator on a variant's orientation in the kinship spectrum and show why calibration averaged over variants can conceal opposing errors. In a new conditional experiment with known covariance, 256 samples, and 970 held-out audit variants, scalar calibration gives an overall rejection probability of 1.00% at a nominal 1% threshold under moderate frequency dispersion, but 1.19% and 0.73% in the lowest and highest structure-loading quintiles. A rank-four spectral correction reduces that spread to 1.010% and 1.005%. Analytic probabilities agree with simulations across 2,000 independent phenotype replicates. These results establish a mechanism under a controlled Gaussian null; they do not validate estimated-covariance scans or reference GWAS programs. We reassess historical simulations, separate environmental mean misspecification from denominator error, and withdraw an external comparison compromised by genotype encoding. A new matched phensim study uses six independent panels per cell, with and without local LD, and verifies every exported genotype. At $n = 800$ with LD, environmental confounding gives official LDAK-KVIK 15.83% rejection at a nominal 1% under the genetic null; adjustment for two genotype PCs reduces this to 0.97%. Corresponding mixmogam KVIK rates are 0.64% and 0.94%. These small-panel stress tests do not establish a general ranking. We implement a HAPNEST-model simulator in phensim and measure generation through 100,000 samples. At 50,000 samples and 20,000 markers, explicit moment-based variance fitting and optional parallel computation give mixmogam median complete-fit times of 33.22 and 32.06 seconds on two fixed panels, versus 25.45 and 30.09 seconds for official LDAK-KVIK. Disabling the genotype cache approximately halves peak resident memory but increases time. Thread changes exceed a prespecified strict numerical tolerance, although the tested significance decisions are unchanged. These workstation timings, with observed system swapping, establish a workload-specific tradeoff rather than general performance or calibration. REML remains the default variance estimator. The research agenda emphasizes independent calibration audits, reference-panel fidelity, error-controlled approximations and measured failure limits.

1 Introduction

A mixed-model association test asks whether a candidate variant explains phenotypic variation beyond specified covariates and a model of genetic background. Its appeal is that the same rela-

tionship matrix can describe close relatives, cryptic relatedness, and broad population structure. Early efficient implementations made this approach practical in model organisms and human association studies (Kang et al., 2008, 2010; Lippert et al., 2011; Zhou and Stephens, 2012). The resulting test is nevertheless conditional on assumptions about the phenotype mean, the background covariance, and the way nuisance parameters are fitted. A precise matrix calculation cannot repair a misspecified mean, and a well-fitted prediction model does not by itself supply a valid association test.

The computational problem has changed with cohort size. Dense methods factorize an $n \times n$ covariance matrix and reuse the factorization across variants. Large-cohort methods instead use iterative fitting, whole-genome prediction, and approximate association statistics. BOLT-LMM combines mixed-model association with a mixture prior on marker effects (Loh et al., 2015); REGENIE constructs whole-genome predictions through successive ridge regressions (Mbatchou et al., 2021); and newer approaches include LDAK-KVIK (Hof and Speed, 2025) and the variational method Quickdraws (Loya et al., 2025). These programs have distinct fitting and testing procedures. A claim about one denominator approximation should therefore be evaluated at the level of that approximation, rather than assigned to all methods with a two-step workflow.

Two statistical choices cut across these computational designs. First, a candidate variant and its correlated neighbours can contribute to the random-effect predictor against which they are tested. This proximal contamination can reduce power; excluding the candidate chromosome or region changes that overlap (Listgarten et al., 2012; Yang et al., 2014). Second, ancestry can influence both allele frequencies and environmental exposures. Modelling genetic covariance and adjusting the phenotype mean with principal components (PCs) address different parts of this problem (Price et al., 2006; Vilhjálmsen and Nordborg, 2013). Neither chromosome exclusion nor covariance modelling makes environmental confounding disappear as a matter of algebra.

This report focuses on a third, more specific issue: the denominator of an association statistic depends on how the tested variant aligns with the covariance eigenvectors. Replacing that denominator by a scalar multiple of the variant norm can work well when this alignment has little effect. Under structure, a satisfactory average can instead hide inflation in one class of variants and deflation in another. The relevant object is the conditional rejection probability for a variant, or a prespecified class of variants, under a stated null model. A single genome-wide median is an incomplete diagnostic of that object.

mixmogam offers a useful setting for studying these distinctions because dense and streaming calculations share genotype handling and several model components. Its earlier work included multi-locus mixed-model association (Segura et al., 2012); the current engine also supplies LOCO and experimental two-step analyses. The present contribution combines a methods and software investigation with a controlled reference-program benchmark. We provide the mathematical contracts needed to interpret its outputs, a new known-covariance experiment with a disjoint set of audit variants, matched phensim simulations with and without population structure and LD, and a critical reading of archived simulations. We then formulate research questions that can distinguish a useful general method from an approximation that works only in a favourable example.

The evidence has several levels. Current software checks assess specified numerical and data-handling contracts. The new experiment isolates calibration geometry under a deliberately simple null. A matched phensim benchmark compares complete scans with the official LDAK-KVIK executable. Larger fixed-panel experiments distinguish estimator choice, numerical parallelism and storage costs. Historical simulations motivate broader questions but predate the scientific correctness review. Table 1 records these boundaries; they apply throughout the report.

Table 1: Evidence used in this revision and the inference it supports. Historical archives are preserved, not silently reclassified as results from the corrected package.

Evidence	Established here	Remaining boundary
Initial review, 2.0.0.dev1	Regression and independent numerical checks; 117 local tests passed; installed-artifact checks	Test coverage does not establish population calibration.
New denominator experiment	Analytic conditional rejection probabilities, checked by independent phenotype replicates	Known Gaussian covariance, small fixed genotype panels, exact leading eigenvectors.
Archived simulations	Descriptive patterns under the recorded historical implementation	New input handling and fitting changes need fresh end-to-end runs.
Matched phensim benchmark	Independent BED decoding and reference-reader checks; null calibration, QTL detection and process resource measurements	Six independent panels per cell; small quantitative-trait simulations, controlled LD and three discrete populations.
Computational extension, frozen 2.0.0.dev3	Projected-moment validation; paired implementation, thread and cache comparisons; 36 complete fits at 50,000 samples and 20,000 markers	Two fixed timing panels; different variance estimators across programs; strict thread-tolerance failures and system-wide swapping retained.
Archived external comparison	A provenance failure was identified: incorrect PLINK export encoding	External-program calibration, agreement, and superiority claims from that archive remain withdrawn.

2 Statistical and computational methods

2.1 Data, estimand, and relationship matrix

Let $\mathbf{y} \in \mathbb{R}^n$ be a quantitative phenotype, $\mathbf{X}$ a full-rank $n \times p$ covariate matrix including an intercept, and $\mathbf{z}_j$ a centered candidate genotype. The coefficient of interest, α_j , is the fixed effect in a specified marginal mixed model after integrating out the random genetic background. It is not automatically a direct causal effect: linkage disequilibrium (LD), omitted exposures, and the choice of background model affect its interpretation. The null is $H_{0j} : \alpha_j = 0$ under that model. A variant without a designated large QTL effect need not satisfy this null if a simulated polygenic background still assigns it an effect.

The relationship matrix summarizes the covariance induced by a polygenic marker model (Yang et al., 2010). For the unweighted matrix, mixmogam centers each marker by its mean among called samples and divides by its empirical standard deviation among those samples. Missing calls become zero after this transformation, corresponding to mean imputation. Write the resulting background matrix as $\mathbf{Z}$, with M retained polymorphic markers. Its raw relationship matrix and a weighted extension are

$$\mathbf{K}_0 = \frac{1}{M} \mathbf{Z} \mathbf{Z}^\top, \quad \mathbf{K}_w = \frac{\sum_{j=1}^M w_j \mathbf{z}_j \mathbf{z}_j^\top}{\sum_{j=1}^M w_j}, \quad w_j \geq 0, \quad \sum_j w_j > 0. \quad (1)$$

Empirical-standard-deviation scaling differs from using the Hardy–Weinberg variance $2f_j(1 - f_j)$; these definitions should not be interchanged in a reference comparison. The optional kinship rescaling subtracts the mean off-diagonal entry from every entry and divides by the resulting mean diagonal. An intercept absorbs the constant-matrix shift in the residual subspace. The raw, unrescaled matrix is used in the new experiment in Section 3.1.

The current input contract is diploid hard calls 0,1,2, with -1 for missing values. It is not a continuous-dosage interface. Minor allele count is measured in allele copies. Sample identifiers must align uniquely; silently intersecting phenotype and genotype rows in different orders changes the scientific problem. PLINK effect-allele orientation, missingness, and identifier preservation are consequently part of the statistical contract, not merely file-format details.

2.2 Gaussian model, variance fitting, and chromosome exclusion

For a candidate in chromosome group g , the working model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{z}_j\alpha_j + \mathbf{u} + \mathbf{e}, \quad \mathbf{u} \sim N(0, \sigma_g^2 \mathbf{K}_{-g}), \quad \mathbf{e} \sim N(0, \sigma_e^2 \mathbf{I}), \quad \mathbf{u} \perp \mathbf{e}. \quad (2)$$

Here $\mathbf{K}_{-g}$ is constructed without the markers in group g . The subscript on σ_g^2 denotes genetic variance, not the group index. With $\delta = \sigma_e^2/\sigma_g^2$, set $\mathbf{H} = \mathbf{K}_{-g} + \delta \mathbf{I}$ and $\mathbf{V} = \sigma_g^2 \mathbf{H}$. This parameterization assumes a positive genetic scale; the zero-genetic-variance limit reduces to a linear model.

The covariance shape is fitted under the null $\alpha_j = 0$. For fixed δ , define the residual precision matrix

$$\mathbf{P}_H = \mathbf{H}^{-1} - \mathbf{H}^{-1} \mathbf{X} (\mathbf{X}^\top \mathbf{H}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{H}^{-1}. \quad (3)$$

Up to constants independent of δ , the profiled restricted log-likelihood is

$$\ell_R(\delta) = -\frac{1}{2} \left\{ \log |\mathbf{H}| + \log |\mathbf{X}^\top \mathbf{H}^{-1} \mathbf{X}| + (n-p) \log \left(\frac{\mathbf{y}^\top \mathbf{P}_H \mathbf{y}}{n-p} \right) \right\}. \quad (4)$$

Thus $\hat{\sigma}_g^2 = \mathbf{y}^\top \mathbf{P}_H \mathbf{y} / (n-p)$ for REML at the fitted δ . Maximum likelihood instead uses n and omits the covariate determinant term. Confusing these objectives changes both the fit and its interpretation. A phenotype entirely explained by $\mathbf{X}$ has zero residual variance and is rejected rather than assigned numerical noise as a variance estimate.

LOCO matrices can be formed by subtracting group-specific cross-products from the genome-wide sum. The dense implementation refits variance components for each group; the streaming infinitesimal path uses one global variance fit and group-specific solves. These are distinct nuisance-fitting procedures. When chromosome labels are numerous, the implementation can merge them into a limited number of groups, so the actual grouping must accompany the results. Exclusion reduces direct overlap between the tested region and its background predictor; it does not imply statistical independence across groups or correct an omitted phenotype mean.

2.3 An explicit moment-estimation option for KVIK

Variance estimation can dominate a streaming fit even after individual matrix products are efficient. KVIK already computes randomized Haseman-Elston (HE) quantities to choose the frequency-weight exponent a , with $w_j = \{f_j(1-f_j)\}^{1+a}$ in Equation (1). The optional `heritability_method="he"` reuses these products to estimate the genetic and residual covariance coefficients. The default remains REML; avoiding its iterative solves and stochastic likelihood calculations changes the estimator, not just its implementation.

Let Q span the covariates orthonormally, $S = \mathbf{I} - QQ^\top$, $d = n - p$, and $K = S\mathbf{K}_w S$. After residualizing and scaling the phenotype, write $\mathbf{y}_s = S\mathbf{y}/\sqrt{\mathbf{y}^\top S\mathbf{y}/d}$. The working covariance is $v_g K + v_e S$. The noise term must be S , not $\mathbf{I}$: projection makes its off-diagonal entries nonzero. Frobenius least squares against $\mathbf{y}_s \mathbf{y}_s^\top$ gives the two moment equations

$$\begin{pmatrix} t_2 & t_1 \\ t_1 & d \end{pmatrix} \begin{pmatrix} v_g \\ v_e \end{pmatrix} = \begin{pmatrix} \mathbf{y}_s^\top K \mathbf{y}_s \\ \mathbf{y}_s^\top \mathbf{y}_s \end{pmatrix}, \quad t_1 = \text{tr } K, \quad t_2 = \text{tr } K^2. \quad (5)$$

Independent Rademacher probes r_b estimate t_2 by $B^{-1} \sum_{b=1}^B \|Kr_b\|^2$, with $B = 32$ by default. The diagonal and products already computed for exponent selection supply the remaining terms. All exponents use the same probes. Exponent selection retains the existing off-diagonal HE score; it is not replaced by the constrained fit in Equation (5).

With $D = t_2 - t_1^2/d > 0$, the unconstrained genetic coefficient is $\{\mathbf{y}_s^\top K \mathbf{y}_s - (t_1/d) \mathbf{y}_s^\top \mathbf{y}_s\}/D$. If either fitted coefficient is nonpositive, the implementation compares the two nonnegative boundary solutions of the same quadratic objective. It does not clip the unconstrained ratio to an arbitrary interval. Nonfinite or numerically unidentifiable moments, including nonpositive curvature, raise an error. The returned diagnostics retain the raw and constrained coefficients, boundary status, D , and the Monte Carlo standard error of t_2 . This last quantity is not a confidence interval for heritability.

The ratio passed to the marker-effect prior is

$$h^2 = \frac{v_g}{v_g + v_e}, \quad h_{\text{projected}}^2 = \frac{v_g t_1/d}{v_g t_1/d + v_e}. \quad (6)$$

The first follows the package’s covariance-coefficient convention; only when $t_1/d = 1$ is it also the mean projected variance fraction. Both are reported. A zero genetic coefficient gives zero polygenic effects. The variational fit retains its existing residual-variance floor of 0.001 on the standardized phenotype scale, including at the opposite boundary; the effective value is reported.

This is a projected single-component moment estimator. It is not LDAK-KVIK’s partitioned HE procedure with large-effect exclusions, nor its optional Monte Carlo REML revision (Hof and Speed, 2025). The pinned reference implementation conditions that revision on sample size and its structure workflow. Matching a command labelled HE therefore does not make the two covariance models or fitted ratios identical.

2.4 What an exact scan makes exact

For a fixed covariance shape $\mathbf{H}$, let $U_j = \mathbf{z}_j^\top \mathbf{P}_H \mathbf{y}$, $q_j = \mathbf{z}_j^\top \mathbf{P}_H \mathbf{z}_j$, and $S_0 = \mathbf{y}^\top \mathbf{P}_H \mathbf{y}$. Generalized least squares gives

$$\hat{\alpha}_j = \frac{U_j}{q_j}, \quad F_j = \frac{U_j^2/q_j}{(S_0 - U_j^2/q_j)/(n - p - 1)}. \quad (7)$$

Under a Gaussian null with this *fixed, correct* covariance shape, $q_j > 0$, and positive residual degrees of freedom, $F_j \sim F_{1, n-p-1}$. In EMMAX-style scans, however, δ is first estimated from the phenotype and then held fixed while variants are tested (Kang et al., 2010). The same algebra can be computed accurately, but the finite-sample reference law is then a plug-in approximation. Throughout this report, “exact scan” refers to the dense calculation with its fitted covariance, not a guarantee of exact type-I error.

The corresponding known-covariance score test is particularly useful for analysis. Replace $\mathbf{H}$ by the fully specified $\mathbf{V}$ in Equation (3) to obtain $\mathbf{P}_V$. Since $\mathbf{P}_V \mathbf{V} \mathbf{P}_V = \mathbf{P}_V$, the null statistic satisfies

$$T_j = \frac{(\mathbf{z}_j^\top \mathbf{P}_V \mathbf{y})^2}{\mathbf{z}_j^\top \mathbf{P}_V \mathbf{z}_j} \sim \chi_1^2. \quad (8)$$

This conditional result is the reference for the new experiment. It avoids conflating an approximation to the denominator with uncertainty in estimated variance components. Variants collinear with the covariates have zero residual norm and are not estimable.

2.5 Scalar calibration and its spectral meaning

Work in orthonormal coordinates spanning the complement of $\mathbf{X}$. Let $R^\top R = \mathbf{I}_d$, $R^\top \mathbf{X} = 0$, and $d = n - p$; use $\tilde{\mathbf{y}} = R^\top \mathbf{y}$, $\tilde{\mathbf{z}}_j = R^\top \mathbf{z}_j$, and $\tilde{\mathbf{K}} = R^\top \mathbf{K} R$. For the rest of this subsection the tildes are suppressed. The infinitesimal solve is $\mathbf{w} = (\mathbf{K} + \delta \mathbf{I})^{-1} \mathbf{y}$. A retrospective statistic uses $D_j^{\text{ret}} = (\mathbf{z}_j^\top \mathbf{z}_j / d) \|\mathbf{w}\|^2$ in the denominator of $(\mathbf{z}_j^\top \mathbf{w})^2$. The matching prospective denominator is $D_j^{\text{pro}} = \sigma_g^2 q_j$, where $q_j = \mathbf{z}_j^\top (\mathbf{K} + \delta \mathbf{I})^{-1} \mathbf{z}_j$. A calibration factor averages $D_j^{\text{pro}} / D_j^{\text{ret}}$ over selected markers. A common factor can reproduce the prospective statistic for all markers only when the relevant ratios are constant.

Write $\mathbf{K} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$ and $a_{\ell j} = \mathbf{U}_\ell^\top \mathbf{z}_j$. The missing geometry is explicit:

$$q_j = \sum_{\ell=1}^d \frac{a_{\ell j}^2}{\lambda_\ell + \delta}, \quad \|\mathbf{z}_j\|^2 = \sum_{\ell=1}^d a_{\ell j}^2. \quad (9)$$

A direction assigned high background variance receives low precision. Two variants with equal norms can therefore have different q_j . A denominator calibrated to the average may be too small for variants concentrated in low-variance directions and too large for variants concentrated in high-variance directions.

For known covariance and a deterministic approximate denominator $\hat{q}_j$, put $r_j = q_j / \hat{q}_j$. The consequence is exact:

$$\tilde{T}_j = r_j T_j, \quad \Pr(\tilde{T}_j > c_\alpha \mid \mathbf{Z}) = \bar{F}_{\chi_1^2} \left(\frac{c_\alpha}{r_j} \right), \quad c_\alpha = F_{\chi_1^2}^{-1}(1 - \alpha). \quad (10)$$

An error that appears modest near the center can matter much more in the tail. Averaging Equation (10) over a variant group gives its expected rejection fraction, without assuming variants are independent. It is a marginal rejection probability, not the family-wise probability of at least one discovery.

2.6 A low-rank correction and its current limits

Retaining the leading k eigenpairs and replacing the rest by their mean $\bar{\lambda}_{>k}$ gives the approximation

$$q_j^{(k)} = \sum_{\ell \leq k} \frac{a_{\ell j}^2}{\lambda_\ell + \delta} + \frac{\|\mathbf{z}_j\|^2 - \sum_{\ell \leq k} a_{\ell j}^2}{\bar{\lambda}_{>k} + \delta}, \quad \hat{q}_j = c_k q_j^{(k)}. \quad (11)$$

The scalar c_k corrects average remaining error; it cannot remove arbitrary variation in the spectral tail. For a LOCO analysis, each approximation should target the covariance used by that group's test. Using a full-genome basis to approximate a different LOCO covariance requires its own error assessment.

The production implementation computes group-specific bases with randomized subspace iteration and currently tries widths 64, 128, 256, and 512. It stops when the coefficient of variation of calibration ratios is at most 3%, subject to the available dimension. The calibration markers also determine the scale, so this stopping criterion is an internal diagnostic. It is not an independent validation and does not bound the error at genome-wide thresholds. The production infinitesimal calibration also selects candidate markers using a phenotype-dependent statistic; this differs from the phenotype-independent selection in our controlled experiment.

For mixture-prior and elastic-net residuals, an infinitesimal spectral factor is transferred before further calibration. Once the residual is a nonlinear, data-adaptive function of $\mathbf{y}$, the proof of

Equation (8) no longer supplies its null law. This transfer is a heuristic. The names “BOLT-style” and “KVIK-style” below denote mixmogam implementations inspired by the corresponding methods; they do not establish equivalence to the reference programs.

2.7 Environmental mean misspecification

Suppose the actual conditional mean includes an omitted exposure vector $\mathbf{d}$ with effect γ : $\mathbb{E}(\mathbf{y} \mid \mathbf{Z}, \mathbf{d}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{d}\gamma$. Even if the covariance $\mathbf{V}$ is known correctly, the numerator in Equation (8) then has mean $\mathbf{z}_j^\top \mathbf{P}_V \mathbf{d}\gamma$. Its distribution becomes

$$T_j \sim \chi_1^2(\eta_j), \quad \eta_j = \frac{(\mathbf{z}_j^\top \mathbf{P}_V \mathbf{d}\gamma)^2}{\mathbf{z}_j^\top \mathbf{P}_V \mathbf{z}_j}. \quad (12)$$

A kinship matrix can assign large variance to an ancestry direction, but it does not automatically place an environmental mean shift in $\mathbf{X}$. Adding covariates that span $\mathbf{d}$ makes the numerator mean zero under this model. Estimated PCs may only approximate that span, and neither the required number of PCs nor a power-preserving adjustment follows from this identity alone. Reviews of structure and mixed models place this distinction in a broader setting (Price et al., 2010; Sul et al., 2018; Yang et al., 2014).

2.8 Numerical implementation and ancillary inference

Dense scans whiten genotype blocks and use matrix products to evaluate Equation (7). Dense factorization costs $O(n^3)$ time and $O(n^2)$ storage, with a dense transformation cost of order n^2m for m tested variants. Streaming methods replace stored kinships by products with standardized genotype blocks. A product with r right-hand sides costs order nMr arithmetic; its memory consists of the right-hand sides, working blocks, and configured caches. Avoiding an $n \times n$ kinship does not mean constant memory: a genotype cache can still approach the size of the standardized data.

Conjugate gradients supplies covariance solves, and stochastic Lanczos quadrature estimates likelihood terms for large samples. Both introduce numerical approximations beyond the statistical model. Association solves now reject nonconvergence. Variational fits expose incomplete convergence, and fit caches respect their options and model ownership. The review also removed an unnecessary LOCO-group axis from temporary products; Section 4.5 reports the measured allocation change with its scope.

Several adjacent routines need equally explicit contracts. The genetic BLUP is $\hat{\mathbf{u}} = \mathbf{K}\mathbf{H}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$; the genetic scale cancels and must not be multiplied in a second time. Default permutation inference operates on orthogonal residual coordinates rather than permuting the correlated entries of a projected residual vector, following the underlying distinction developed by Abney (2015). Exact exchangeability requires the stipulated Gaussian covariance; substituting fitted variance components still requires calibration checks. Finally, the normal-approximation Bayes factor requires effect estimates, standard errors, and an explicit effect-variance prior (Wakefield, 2009). It is not a replacement for a joint fine-mapping model. Appendix A gives these identities and their interpretation.

2.9 Reuse, bounded workspaces and optional parallelism

The computational changes first remove work whose result is already available. Variational fitting caches only requested fold-specific Gram matrices, reuses full-block Gram matrices for LOCO, and reuses final predictions. A persistent matrix-product workspace replaces repeated allocations.

Residual updates retain float64 accumulation while preparing the next matrix-product input in the genotype storage precision, float32 by default. This distinction matters: converting the entire calculation to float32 would be a different numerical contract.

Temporary widening is bounded without dividing the reduction axis: genotype standardization uses variant tiles with a 64 MiB scratch target; prediction, HE diagonal terms and retrospective statistics use 16 MiB targets. These are explicit temporary-workspace budgets, not process RSS limits. Genotype validation and PLINK reading also use bounded chunks; reading no longer retains an entire packed BED beside its decoded copy. The stored input is still an int8 hard-call matrix, not a packed or fully out-of-core genotype backend.

With `n_threads>1`, an optional Numba kernel prepares independent variants in parallel. Each worker computes called-sample means, a two-pass float64 variance and the covariate projection before storing float32 values. Sample reductions remain serial within a variant, with no fast-math reassociation. The initial scratch is $8(n+p)$ bytes per active worker, separate from outputs. Retaining Mp float64 projection coefficients permits later decoding directly from int8 calls and prepared moments. On the uncached variational path, at most 128 markers are decoded per subblock. Other stages may still materialize their configured parent blocks, so this is a bounded fitting workspace rather than constant total memory. The one-thread default retains the NumPy preparation path.

Coordinate sweeps parallelize independent candidate models and LOCO fits while preserving the Gauss–Seidel marker order within each model. For sufficiently large multi-model fits, residual matrix products also use an output-row partition and a pool capped at four workers. Each output entry retains its complete marker reduction; there is no shared writable sum across workers. Detected BLAS pools are temporarily limited to one, and thread settings and workers are restored on exit. Apple Accelerate requires an external thread limit because it is not detected by `threadpoolctl`. Parallel preparation and BLAS reductions can still change floating-point rounding; a scientifically unchanged algorithm is not a promise of bitwise equality.

The standardized-genotype cache has a default budget of 4×10^9 bytes and is retained when its size is less than or equal to that budget. Consequently a float32 cache at $n = 50,000$, $M = 20,000$ lies exactly on the inclusion boundary. Disabling it saves retained storage but repeats decoding during fitting. The comparison in Section 4.8 measures this exchange in complete fits rather than inferring it from a decoder microbenchmark.

3 Evaluation design and provenance

3.1 New conditional experiment with known covariance

The new experiment isolates Equation (10); it does not call an end-to-end GWAS scan. We generate 256 diploid samples in four equal populations, with 2,000 background and 1,000 candidate markers. For each marker, the ancestral frequency f_j is uniform on $[0.1, 0.9]$. Population-specific frequencies are independent draws from $\text{Beta}(f_j(1/\theta - 1), (1 - f_j)(1/\theta - 1))$ for $\theta > 0$; at $\theta = 0$ they all equal f_j . Hard calls are binomial with two trials. The values $\theta \in \{0, 0.05, 0.20\}$ control frequency dispersion. We do not report them as measured F_{ST} , whose estimation and interpretation require a separate definition (Bhatia et al., 2013).

Markers are independent conditional on population frequencies and sample membership. Population structure nevertheless correlates their sample-level patterns. Monomorphic markers are removed: one background marker is dropped in the $\theta = 0.20$ panel; all 1,000 candidates remain in every panel. The package’s raw relationship matrix is checked against a direct standardized cross-product. We remove an intercept in orthonormal residual coordinates, set $\delta = 1$ and genetic

scale one, and use a full eigendecomposition to obtain the known covariance. Direct linear solves and spectral denominators agree to numerical precision.

Thirty randomly selected candidates calibrate each approximation; the other 970 are held out for audit. Selection is independent of phenotype. The scalar denominator is the variant norm times the mean $q_j/\|\mathbf{z}_j\|^2$ among calibration markers. For the spectral method, we try $k \in \{4, 8, 16, 32, 64, 128\}$ and select the first width whose calibration-ratio coefficient of variation is at most 3%. All three panels select $k = 4$. These exact leading eigenvectors provide a favourable test of spectral geometry, not a validation of the production randomized eigensolver or its width-selection procedure.

Audit variants are divided into quintiles by the share of their squared norm explained by the top three background eigenvectors. For each method and quintile we calculate the analytic mean rejection probability at $\alpha = 0.01$ and 5×10^{-8} . We also simulate 2,000 independent Gaussian phenotype replicates per fixed panel and compare the empirical rejection fraction at 1% with its analytic value. Methods use the same phenotypes. Monte Carlo standard errors are calculated from the replicate-level fractions, retaining correlation among variants within each replicate. We do not estimate the extreme-tail probability by counting a handful of simulated exceedances.

The three genotype panels use seeds 20261003–20261005. Consequently, differences across θ also include differences among three realized panels; there are no genotype-replicate confidence intervals. The analytic result is exact conditional on each realized panel and its fixed approximation. It is not an estimate of an average over all possible populations.

3.2 Matched phensim benchmark with official LDAK-KVIK

We froze a factorial protocol before examining association outcomes and used an excluded pilot to check interoperability and resource demands. Each cell has six independent genotype and phenotype replicates, $n = 800$ diploid samples, and $m = 6,000$ markers on six equal chromosomes. All genotypes and phenotypes are generated by phensim. Its `simulate_population_structure` function uses three populations with uniformly sampled membership and Balding–Nichols frequency drift. The software’s `fst` parameter is zero or 0.05; it controls frequency dispersion and is not an empirical estimate of F_{ST} . Shared ancestral frequencies are $0.3 + N(0, 0.05^2)$, clipped to $[0.05, 0.95]$; population frequencies are clipped to $[0.01, 0.99]$.

Two LD settings isolate local linkage from ancestry correlation. Markers are either independent conditional on population or generated in 50-marker blocks using two latent Gaussian AR(1) haplotypes per person with correlation parameter 0.8. Population-specific thresholds convert each haplotype to allele indicators. Blocks are independent conditional on population, but pooled genotypes share ancestry correlations across blocks and chromosomes. We added the optional block generator within phensim, preserving its existing independent-marker seeded draws. This controlled LD experiment does not reconstruct human demographic history. Markers with pooled minor-allele frequency below 0.01 are removed before either program receives them; calls are complete.

The base global-null trait has no genetic effects and Gaussian residuals. The mixed trait has target genetic variance 0.5, divided equally between an infinitesimal background and ten normally distributed QTL effects. Both components use chromosomes 1–5 only, with QTLs in distinct blocks; chromosome 6 has no generating effects. Shared effects across populations are assumed. Phensim records every component, causal index and effect. For environmental confounding we pass population labels $-1, 0, 1$ as an explicit exposure E , standardized within the sample, and form

$$y = \text{standardize} \left\{ \sqrt{s}E + \sqrt{1-s}(u + q + e) \right\}, \quad s \in \{0, 0.2\}. \quad (13)$$

The component targets are not asserted to equal realized variance fractions: population structure can correlate E , u and q . The archive therefore includes their covariance matrix.

Four analysis cells comprise unstructured data, structured data without environmental effects, structured data with the exposure, and the same confounded phenotype adjusted for the two leading genotype PCs. PCs are computed without phenotypes and supplied identically to all methods. Every analysis includes an intercept. Exact LOCO EMMAX, mixmogam BOLT-inf, mixmogam KVIK and official LDAK-KVIK use their default fitting and calibration settings with explicit seeds and one thread. These are comparisons of supplied procedures, whose priors and tuning searches differ; matching inputs does not make their statistical models identical. Here “mixmogam KVIK” denotes this package’s Python/Numba implementation of a KVIK-style procedure, while “LDAK-KVIK” denotes the official standalone executable. The distinction includes statistical choices, such as variance-component estimation, as well as implementation language. The official documentation states that LDAK-KVIK has primarily been tested above 50,000 samples, while also supporting smaller datasets with potentially less benefit from LOCO prediction.¹ The primary panels are therefore small-sample stress tests, not a proxy for performance in a densely genotyped biobank.

Phensim writes PLINK calls counting BIM allele A2. An independent decoder checks every BED call against the simulated matrix; mixmogam must read the complementary A1 counts. LDAK’s own statistics command verifies the same marker IDs, sample IDs, allele labels, allele means and call rates. Both analysis programs read the same BED/BIM/FAM files and serialized phenotypes and covariates. The official native Mac executable `ldak6.3.mac` is pinned to repository revision `995e18753a0a`, with its full URL and SHA256 in the archive; the generic “Version 6” banner alone does not identify this build.

Calibration is measured on all markers under the global genetic null, and separately on chromosome 6 under the mixed trait. The latter shares ancestry with causal chromosomes, so rejection assesses control of ancestry associations as well as statistical approximation. Other markers are not automatically labelled null simply because they are not large QTLs. We report rejection at 0.05, 0.01 and 0.001, genomic inflation from p-equivalent chi-square statistics, and population-loading, frequency and local-LD quintiles. Loading is the fraction of genotype sum of squares explained by population membership, chosen without phenotype outcomes. Power is direct causal-marker detection at $0.05/m$ and 5×10^{-8} . Monte Carlo standard errors use independent replicate panels, not the number of correlated markers. Six replicates cannot establish calibration at genome-wide significance thresholds.

Each method runs in a separate process. Wall time includes startup, input loading, fitting, scanning and result writing; LDAK-KVIK time sums its two stages. Kernel `wait4` counters provide peak resident memory, using the larger stage peak for LDAK-KVIK. Common simulation, export validation and PC construction are recorded separately. Method order rotates across replicates, code caches may be warm, and the macOS power guard requires AC power with Low Power Mode disabled. This is a bounded laptop comparison, not a biobank-scale performance claim.

3.3 Larger workloads and reference-based simulation

After inspecting the primary study we fixed a descriptive runtime extension at $n=2,000$ with $m=12,000$ and $n=4,000$ with $m=24,000$, using two independent replicates for the unstructured and PC-adjusted confounded mixed traits. All four methods retain their fitting defaults. Increasing n and m together describes a workload curve, not an isolated complexity exponent. These panels are not pooled into the six-replicate calibration experiment.

For larger n , we implemented the genotype-copying model of HAPNEST (Wharrie et al., 2023)

¹LDAK-KVIK [recommendations](#), accessed 3 October 2026.

in phensim. A phased reference is expanded by copying segments whose ages and lengths satisfy

$$T \sim \text{Gamma}(2, \text{scale} = N_e/N_{\text{ref}}), \quad L \mid T \sim \text{Exp}(\text{rate} = 2T\rho). \quad (14)$$

Here N_{ref} counts reference diploids, L is in centimorgans, and T is in generations. Donors are uniform within the assigned population; an allele is retained only if its mutation age exceeds T . Chromosome boundaries restart copying. The independently written implementation follows upstream phase and endpoint conventions, with a fused Numba kernel and a NumPy oracle. It does not implement HAPNEST’s ABC parameter tuning, demographic admixture, or upstream phenotype model.

The simulator resource experiment uses 600 phensim reference diploids, 12,000 markers, three frequency-drift populations, latent AR(1) correlation 0.8, map spacing 0.001 cM, mutation ages 1,000 generations, $N_e = 10,000$ and $\rho = 0.7$. These are controlled inputs, not estimates from human data. Three seeded repeats compare backends at 2,000 samples and assess Numba through 100,000 samples. At 50,000 samples, output is streamed, retained in RAM, or memory-mapped. Matched modes must give identical output hashes. First-call compilation or cache loading is timed separately. Warm generation includes batch checksums; both OS process peak RSS and a 2 ms sampled peak are recorded. Memory-mapped file pages can become resident: only streaming avoids materializing the entire result.

The association extension uses $n=10,000$ and 50,000, $m=12,000$, and one mixed-trait realization per unstructured or PC-adjusted confounded cell. The unstructured control pools reference donors; the structured cell restricts donors by population. Setting $N_e = 50N_{\text{ref}}$ matches their segment-age law. We retain the primary chromosome-6 genetic null. Tiled phenotype generation preserves the dense GRM factor and Gaussian innovations up to summation roundoff. Genotype PCs use an iterative eigensolver with audited residuals, never an $n \times n$ matrix. Separate preparation processes keep simulation arrays outside association RSS. One realization describes feasibility and costs, not stable power or tail calibration; a synthetic reference does not establish human-data fidelity.

3.4 Separating estimator validation from the 20,000-marker workload

We evaluated projected HE before using it for the larger workload. Four independent phensim genotype panels contain $n = 800$, $M = 600$: two unstructured panels and two with frequency-drift parameter 0.08 and two genotype PCs. For each covariance-coefficient target $h^2 \in \{0, 0.3, 0.6\}$, 24 Gaussian phenotypes per panel give 288 draws. The primary comparison fixes $a = -1$ to match the generating covariance and compares 32-probe HE with an independent dense solution of Equation (5). Components are not rescaled to force their realized sample variances. A separate 64-bit experiment compares complete HE and REML scans on the first eight global-null phenotypes per panel. These checks concern a specified covariance and Gaussian null; they do not validate adaptive exponent selection or confounded positive-heritability association scans.

The 20,000-marker extension uses two new HAPNEST-model panels at $n = 50,000$: unstructured, and structured with the environmental exposure in Equation (13) plus two genotype PCs. It retains the six-chromosome, ten-QTL and polygenic-background design of Section 3.3, with 600 simulated reference diploids, AR(1) parameter 0.8, pooled or population-restricted copying, and $N_e = 50N_{\text{ref}}$. From 20,400 candidates, a deterministic genotype-only rule retains the first MAF-passing markers up to chromosome quotas (3350, 3350, 3350, 3350, 3300, 3300). Selection precedes PCs, QTL selection and phenotype generation; original candidate indices are retained. Both panels have zero MAF exclusions and 400 quota removals. Chromosome 6 therefore contains 3,300 genetic-null markers. Six groups preserve the earlier LOCO workload structure, although these are new draws rather than nested subsets of the 12,000-marker panels.

An independent export audit checks every call, allele orientation and sample/marker order; reference-program statistics check the same inputs. PCA uses bounded-memory products and audited eigenpair residuals. All component covariances are retained, since the nominal trait-component targets need not equal realized variance fractions. Common preparation and validation run separately from association processes.

The prespecified primary design contains 24 complete fits: two methods, two requested thread counts (one and four), three timing repetitions and two fixed panels. A separate 12-fit experiment compares cache budgets 4×10^9 and zero bytes with four mixmogam workers. Every local fit explicitly selects HE; the model grid and convergence settings are held fixed. Method-first order and thread positions are balanced; cache-first order is balanced across the six pairs. Each fit starts in a fresh process, with tiny compilation/cache warm-ups excluded. Local time includes startup, imports, input, fitting and output. Official time sums its two native stages and excludes subsequent parsing and compression; RSS is their larger stage peak. Thus timing boundaries are explicit but not perfectly symmetric.

All runs use the 16-GB Apple M2 Pro, AC power and Low Power Mode disabled. Local BLAS/OpenMP environment limits are one with a Numba pool ceiling of eight; `n_threads` requests one or four workers. The official Mac executable receives the corresponding environment limit and native `--max-threads` argument. Requested counts do not assert identical thread execution across implementations. The fitted data, drivers and package code are frozen; the timed package identifies itself as 2.0.0.dev3. Repeated timings on two phenotypes measure computational variability, not biological replication. The new data are also distinct from the small known-covariance HE validation panels.

3.5 Historical structure and environmental experiments

The archived structure experiment used 1,300 samples, 20,000 markers, and five chromosomes, with both unstructured genotypes and a four-population generator labelled $F_{ST} = 0.3$ in the archive. We retain that label as a generator setting, not a newly estimated differentiation measure. Six phenotype replicates shared each genotype panel. Genetic signal was generated outside the last chromosome from a polygenic component of variance share 0.4 and ten large QTLs of variance share 0.1. The last chromosome has no generating marker effects, making it useful for a null-chromosome diagnostic. Sparse QTLs, realized cross-chromosome correlations, and variance estimation mean that this is not the known-Gaussian-covariance experiment above. Loading quintiles use the leading ten kinship eigenvectors.

The archived coalescent experiments used mutation and recombination rates 10^{-8} per base pair per generation, effective population size 10^4 , and minor allele frequency above 1% (Kelleher et al., 2016; Baumdicker et al., 2022). Scenario S2 contains two demes separated 400 generations earlier, 400 diploids per deme, and 100 independently simulated blocks of 200 markers. Its environmental exposure is the standardized deme indicator. It is generated from population membership rather than constructed as a PC of the tested genotypes; it is nevertheless strongly correlated with genotype. The raw trait and standardized exposure are each multiplied by $\sqrt{1/2}$, added, and then standardized together. The original genetic variance target of 0.25 becomes 0.125 before final standardization; the environmental variance target is 0.5. Realized component variances and cross-covariances can change their shares of the final phenotype variance. This is a strong quantitative-trait stress test, not a binary liability model or a typical confounding level.

We report S2 statistics in blocks without a designated large QTL. Those blocks still contain infinitesimal effects. We therefore call them *QTL-free blocks*, not strict-null blocks, and do not interpret their detection count as a formal false-positive rate. The same qualification applies to S1

and S3, which have $n = 800$ and $4,000$, $m = 20,000$ and $50,000$, and 20 and 30 large QTLs, respectively. Their target genetic variance share is 0.5, divided equally between a polygenic background and large QTLs. Each scenario has only three independent simulation seeds.

3.6 Summaries, exclusions, and reproducibility

Genomic inflation is summarized as the median chi-square statistic divided by the median of χ_1^2 , approximately 0.455. This statistic mixes model error, true signal, and sampling variation; it is not an estimand of confounding alone. Loading-stratified rejection fractions are more informative about the denominator hypothesis. For the six historical phenotype replicates, Figure 3 uses Student- t intervals for the mean with five degrees of freedom. These intervals concern variation among phenotypes conditional on a fixed genotype panel; six replicates offer limited precision. S2 displays all three seeds as well as their means.

The archived comparison with the external LDAK program used a faulty PLINK encoding path. A matched-input claim cannot survive that failure. We exclude its empirical results, concordance coefficients, and real-data calibration curves from this revision. The underlying RegMap resource (Horton et al., 2012) remains suitable for a future controlled benchmark, but neither its archived export nor a redraw of its old results validates the corrected package.

New numerical results record the generating script, seeds, software versions, requested BLAS thread limits, genotype hashes, and package baseline commit. Figure generation records hashes of its actual inputs and outputs. Historical archives retain their original source snapshots; we do not attach the current version number to computations performed by an earlier implementation.

4 Results

4.1 A nearly correct average can conceal conditional error

Table 2 and Figure 1 show the new conditional experiment. With no frequency differentiation, the scalar approximation is close to the known-covariance test: its analytic rejection probability is 0.99% overall and 1.00% versus 0.97% in the extreme loading quintiles. At $\theta = 0.05$, the overall value remains 1.00% after rounding, but the extreme quintiles separate to 1.19% and 0.73%. The average has become reassuring for the wrong reason: deviations in opposite directions largely cancel.

At $\theta = 0.20$, the scalar approximation rejects 1.25% overall, with 2.21% in the lowest loading quintile and 0.31% in the highest. The corresponding analytic rejection probabilities at a nominal 5×10^{-8} are 27.4 and 0.025 times nominal. These tail ratios follow from the fully specified Gaussian model and Equation (10); they are not empirical estimates of real-cohort false-positive rates. They demonstrate why a denominator should be assessed at the intended testing threshold.

The rank-four correction substantially reduces the loading gradient in these panels. At $\theta = 0.05$, its extreme-quintile probabilities are 1.010% and 1.005%; at $\theta = 0.20$, they are 0.998% and 0.992%. Its overall analytic rejection ratios at 5×10^{-8} range from 0.97 to 1.06 across the three panels. Held-out root-mean-square relative denominator-ratio errors are 1.36–1.53%. This is evidence that the leading structure directions explain most of the approximation error in these particular designs. It neither proves that four directions suffice elsewhere nor tests the accuracy of an estimated leading subspace.

The empirical rejection fractions agree with their analytic targets within the script’s stated numerical and Monte Carlo tolerance. For the exact denominator, the overall empirical fractions range from 1.006% to 1.007%, with standard errors approximately 0.010 percentage points. There

Table 2: New known-covariance experiment. Rejection probabilities at a nominal 1% threshold, expressed as percentages. “Simulated” is the overall mean $\pm$ one Monte Carlo standard error across 2,000 independent phenotype replicates. q1 and q5 are analytic probabilities in the extreme loading quintiles among 970 held-out audit variants.

θ	Denominator	All audit variants		Analytic	
		Analytic	Simulated	q1	q5
0	Exact	1.00	1.01 ± 0.01	1.00	1.00
0	Scalar	0.99	1.00 ± 0.01	1.00	0.97
0	Spectral, rank 4	0.99	0.99 ± 0.01	0.99	0.99
0.05	Exact	1.00	1.01 ± 0.01	1.00	1.00
0.05	Scalar	1.00	1.01 ± 0.01	1.19	0.73
0.05	Spectral, rank 4	1.01	1.02 ± 0.01	1.01	1.01
0.2	Exact	1.00	1.01 ± 0.01	1.00	1.00
0.2	Scalar	1.25	1.26 ± 0.01	2.21	0.31
0.2	Spectral, rank 4	1.00	1.00 ± 0.01	1.00	0.99

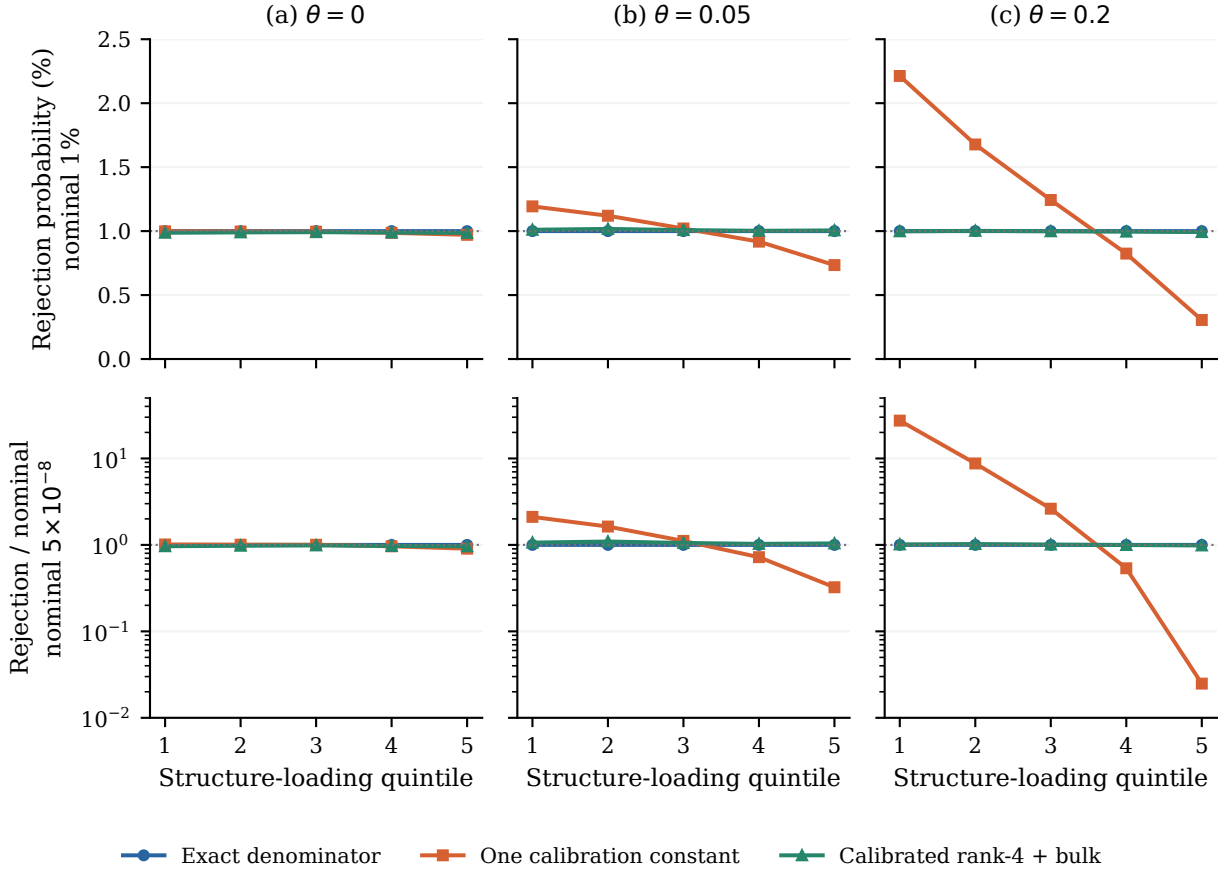


Figure 1: Conditional calibration under known covariance. The top row shows analytic rejection probabilities at a nominal 1%; the bottom row shows the ratio of analytic to nominal rejection probability at 5×10^{-8} on a log scale. Each column is one fixed genotype panel. Candidate variants are held out from approximation calibration and ordered by loading on the leading three background eigenvectors. The spectral method uses four exact eigenvectors and a flat bulk. Columns share vertical scales. Panels describe denominator geometry, not fitted-covariance or external-program performance.

is no fitting uncertainty in this comparison: a score with an exact denominator must have the stipulated marginal law. The simulation checks the implementation of that construction and quantifies the sampling variation of the reported summaries.

4.2 Matched reference comparisons depend on mean adjustment

All 384 planned method runs returned valid p-values for every retained variant. Execution is distinct from convergence: mixmogam KVIK reported incomplete cross-validation in six unadjusted confounded mixed-trait runs, two of which also had incomplete LOCO fits. These remain in the summaries. An independently regenerated structured/PC-adjusted case reproduced the serialized inputs and all four methods' p-values exactly using frozen source snapshots.

Figure 2 separates genetic-null rejection, mixed-trait rejection on chromosome 6, and causal-marker detection. In unstructured global-null cells, mean rejection at 1% is approximately 1.00–1.06% across methods and LD settings. Under structure and the mixed trait, LDK-KVIK's null-chromosome rejection is 1.87% without local LD and 1.82% with LD. This assesses the complete procedure, not an isolated denominator approximation.

Omitting the environmental mean shift produces the largest disagreement. With LD, LDK-KVIK gives $\lambda_{GC} = 3.455$ and 15.83% genetic-null rejection at 1%; exact LOCO and mixmogam BOLT-inf give 1.34% and 1.29%, and mixmogam KVIK gives 0.64% (Table 3). LDK-KVIK rejection is 21.5–25.2% in four panels and 1.52–1.82% in two. Some inflated runs reach LDK-KVIK's lower heritability boundary after revision of the initial estimate. This motivates a diagnostic experiment; it does not identify the cause. The mixmogam deflation also fails nominal calibration.

Adding the same two PCs reduces LD global-null rejection to 0.94% (mixmogam KVIK) and 0.97% (LDK-KVIK). Mixed-trait null-chromosome rejection is 1.13% and 1.05%. Causal-marker detection is 48.33% and 50.00%; the paired mixmogam-minus-LDK difference is -1.67 percentage points, with a six-panel Student-t interval of $[-5.95, 2.62]$. A power difference is unresolved. Detection gains in inflated, unadjusted cells must not be interpreted as calibrated power. Explicit mean adjustment materially changes the comparison in these panels; neither a genome-wide median nor one favourable cell establishes general equivalence.

4.3 Historical scans show a related pattern, with weaker identification

The archived strong-structure scan exhibits the same qualitative gradient (Table 4, Figure 3). For the scalar infinitesimal implementation, overall λ_{GC} is 0.97, but the lowest and highest loading quintiles have $\lambda_{GC} = 1.34$ and 0.64, and rejection fractions of 2.45% and 0.11% at 1%. The historical spectral correction gives 1.14% and 1.06% in those quintiles, close to the dense LOCO results of 1.12% and 1.11%. The KVIK-style implementation retains a pronounced gradient.

These results are supportive diagnostics, not a second proof of the mechanism. Variance fitting, sparse generating effects, calibration-marker selection, and the earlier implementation all contribute to the final statistics. The new experiment is stronger for identifying denominator error precisely because it holds those features fixed or removes them. Conversely, its simplicity limits what it says about the complete scan. Both evidence types are useful when their roles are kept distinct.

4.4 Covariance adjustment and environmental adjustment are different

In the historical S2 stress test, the leading PC predicts deme membership with squared correlation approximately 0.97. Without kinship adjustment, QTL-free blocks have $\lambda_{GC} = 11.90$; the mean statistic among variants with the highest environmental loading is 63.24. Dense LOCO reduces these summaries to 1.24 and 2.27, but substantial residual structure remains (Table 5, Figure 4).

Including PC1 reduces them further to 1.09 and 1.13. The corresponding all-variant λ_{GC} falls from 1.27 to 1.11. The residual is reduced, not demonstrably eliminated.

The non-LOCO scan has all-variant $\lambda_{GC} = 0.98$ and QTL-free $\lambda_{GC} = 0.96$, yet its upper-loading mean remains 1.77 and its locus power is lower than LOCO’s (0.13 versus 0.20). Apparent aggregate calibration is therefore compatible with both residual environmental structure and signal attenuation. These data do not identify their separate causal contributions; a factorial simulation holding the mean and covariance mechanisms apart is needed.

A related interpretation problem appears in S1 and S3. Replacing fixed 200-marker intervals with LD-aware boundaries reduces the number of QTL-free detections. In the recorded paired checks, dense LOCO’s mean count changes from 1.33 to zero across three S1 seeds; the single paired S3 check changes from 17 to 11. This is consistent with large-QTL signal crossing arbitrary boundaries. It does not establish that every remaining detection is an error: the background is polygenic, and alternative block definitions also change the unit of discovery. Methods for constructing approximately independent LD blocks are useful here (Berisa and Pickrell, 2016; Privé, 2022), but evaluation must still define which biological or statistical null a “false locus” is meant to represent.

4.5 Correctness and computational evidence

The initial correctness review recorded 117 passing local tests. Its installed wheel check on Python 3.10 with NumPy 1.26.4 and SciPy 1.11.4 passed 71 tests, with five skips for unavailable optional HDF5/plotting components. These checks include independent numerical references and reproductions of specific defects. The corrected contracts cover PLINK coding, sample alignment and allele counts, BLUP scaling, variance fitting, weighted kinships, window complements, permutation geometry, and effect/standard-error export. Passing them supports those contracts; it cannot substitute for an end-to-end calibration study.

One allocation change has a directly recorded before/after comparison. For 800 samples, 1,000 markers, ten LOCO groups, and 80 right-hand sides, five warm matrix-product repetitions gave identical output arrays. The measured peak allocation fell from 7,235,824 to 2,625,120 bytes, a 64% reduction. This measurement uses Python allocation tracing and excludes the pre-existing genotype cache and native BLAS allocations; it is not whole-process peak resident memory. The short timings do not support a general runtime-speedup claim.

Older timing archives establish why batching and streaming are useful, but not current competitive performance. One internal comparison recorded 0.27 seconds for a batched EMMAX scan versus 33.7 seconds for a per-variant least-squares reference at $n = 2,000$, $m = 10,000$ (about 123-fold). Historical S3 times were 221 seconds for 25-group dense LOCO and 18 seconds for the infinitesimal two-step path; a 2.7-second non-LOCO scan excluded its preceding fit. These workloads differ in both statistical procedure and accounted work. They were measured on a development laptop under concurrent activity, not a controlled cross-program benchmark.

At $n = 10,000$, a historical stochastic-Lanczos fit obtained $\delta = 0.91724$ versus 0.91639 for a full fit, a 0.09% difference in one case. That agreement concerns one nuisance estimate. A historical scan based on a truncated spectrum still produced $\lambda_{GC} = 1.30$, versus 0.85 for the full non-LOCO scan. Neither value establishes nominal calibration in that signal-containing simulation. Accurate likelihood approximation, accurate association denominators, and calibrated tests are separate requirements.

Table 3: LD panels in the matched comparison. Rates are percentages; $\pm$ is one Monte Carlo standard error across six panels. Inflation uses the global genetic null. Rejection columns use that null and the mixed trait’s null chromosome. Detection uses $0.05/m$.

Method	λ_{GC}	Global null	Null chromosome	Detection
<i>Unstructured</i>				
Exact LOCO	1.012	1.03 ± 0.07	0.83 ± 0.15	46.67 ± 3.33
mixmogam BOLT-inf	1.016	1.03 ± 0.08	0.88 ± 0.14	45.00 ± 3.42
mixmogam KVIK	1.013	1.01 ± 0.08	0.97 ± 0.14	43.33 ± 3.33
LDAK-KVIK	1.018	1.06 ± 0.08	1.05 ± 0.14	50.00 ± 3.65
<i>Structured</i>				
Exact LOCO	0.987	0.91 ± 0.07	1.00 ± 0.20	41.67 ± 3.07
mixmogam BOLT-inf	0.995	0.92 ± 0.06	1.20 ± 0.21	41.67 ± 4.01
mixmogam KVIK	0.997	0.92 ± 0.06	1.15 ± 0.16	48.33 ± 3.07
LDAK-KVIK	0.983	0.93 ± 0.06	1.82 ± 0.77	48.33 ± 3.07
<i>Structured + environment</i>				
Exact LOCO	1.115	1.34 ± 0.07	1.08 ± 0.20	38.33 ± 4.01
mixmogam BOLT-inf	1.121	1.29 ± 0.07	1.38 ± 0.19	38.33 ± 4.01
mixmogam KVIK	0.945	0.64 ± 0.05	0.68 ± 0.22	33.33 ± 5.58
LDAK-KVIK	3.455	15.83 ± 4.51	6.27 ± 4.77	40.00 ± 3.65
<i>Same trait + 2 PCs</i>				
Exact LOCO	1.009	0.94 ± 0.07	1.00 ± 0.19	40.00 ± 3.65
mixmogam BOLT-inf	1.008	0.95 ± 0.06	1.17 ± 0.25	40.00 ± 3.65
mixmogam KVIK	1.010	0.94 ± 0.06	1.13 ± 0.16	48.33 ± 4.01
LDAK-KVIK	0.940	0.97 ± 0.06	1.05 ± 0.16	50.00 ± 3.65

Table 4: Historical simulated strong-structure panel, six phenotype replicates. q1 and q5 denote loading quintiles. Rejection fractions are for the chromosome without generating effects, under the qualifications in Section 3.5. These are archived implementation results, not a rerun of version 2.0.0.dev1 or reference programs.

Method	$\lambda_{GC}, \text{ all}$	λ_{GC}		Rejection (%)	
		q1	q5	q1	q5
Exact LOCO	1.00	1.05	1.02	1.12	1.11
Infinitesimal, constant	0.97	1.34	0.64	2.45	0.11
Infinitesimal, spectral	1.00	1.05	1.01	1.14	1.06
KVIK-style	1.02	1.37	0.73	2.92	0.11

Table 5: Historical S2 environmental stress test, means over three seeds. Loading refers to squared correlation with the deme exposure. Detection counts are significant QTL-free blocks, not false positives under a strict null; these blocks also contain polygenic background effects. Power is the fraction of designated QTL-containing loci detected under the archived Bonferroni rule.

Method	$\lambda_{GC}, \text{ QTL-free}$	Mean χ^2		QTL-free detections	Locus power
		Bottom half	Top 1%		
Linear model	11.90	2.20	63.24	90.33	1.00
LMM, no LOCO	0.96	0.87	1.77	0.00	0.13
Exact LOCO	1.24	1.13	2.27	0.67	0.20
Exact LOCO + PC1	1.09	1.10	1.13	0.67	0.20

4.6 Measured time, resident memory, and simulation scale

New timings use an Apple M2 Pro with 16 GB RAM, Python 3.14.6, NumPy 2.4.6 and Numba 0.66.0, with one thread per method. mixmogam KVIK uses its compiled kernels. Across the 96 primary inputs its median time is 2.57 seconds versus 0.74 seconds for LDAK-KVIK, with median RSS 319.5 versus 40.1 MiB (Table 6). The approximately 3.5-fold ratio of medians is workload-specific; startup and loading matter at this size. The native Mac LDAK build lacks MKL, so this is not a comparison of each program’s fastest available hardware build.

Figure 5 and Table 7 extend the controlled LD workloads to 4,000 samples and 24,000 markers. Dense exact scans become substantially more expensive; LDAK-KVIK retains lower observed time and RSS than mixmogam KVIK. Because n and m both increase, the slopes do not identify a sample-size complexity exponent.

Table 6: Primary end-to-end resources at $n=800$ and $m=6,000$. Each row covers the same 96 inputs. LDAK-KVIK time sums both stages; its RSS is the larger stage peak. Common data preparation is excluded.

Method	Runs	Median s	Range s	Median MiB	Max MiB
Exact LOCO	96	2.58	1.96–3.65	466.9	504.3
mixmogam BOLT-inf	96	1.95	1.51–2.80	277.5	311.7
mixmogam KVIK	96	2.57	1.65–4.45	319.5	354.5
LDAK-KVIK	96	0.74	0.25–1.93	40.1	45.6

Table 7: Median seconds for the workload extension and maximum peak RSS at the largest workload. Mixed traits have latent LD correlation 0.8. Column labels show samples/markers; preparation is excluded.

Method	800/6,000	2,000/12,000	4,000/24,000	Max MiB
<i>Unstructured</i>				
Exact LOCO	2.57	12.22	91.41	1291
mixmogam BOLT-inf	2.07	4.60	15.97	1065
mixmogam KVIK	2.76	5.02	15.77	1089
LDAK-KVIK	0.72	1.39	3.44	154
<i>Structure + environment + 2 PCs</i>				
Exact LOCO	2.81	9.77	94.11	1360
mixmogam BOLT-inf	2.46	4.37	14.18	1090
mixmogam KVIK	2.84	5.28	19.08	1177
LDAK-KVIK	0.84	1.46	5.33	153

The HAPNEST-model kernel supplies larger cohorts without generating a full ancestral graph. At 2,000 samples and 12,000 variants, median warm generation time falls from 22.28 seconds for NumPy to 0.48 seconds for Numba, a 46.8-fold ratio; all paired output hashes agree. Streaming 100,000 samples takes 25.73–28.40 seconds and at most 198.6 MiB process RSS across three repeats (Table 8). These times include checksums but exclude loading the small reference and first-call setup. They do not compare against the upstream Julia program. At 50,000 samples a full int8 matrix requires 600 million bytes; RAM and memory-mapped modes retain much more resident memory than streaming. Disk backing alone does not guarantee low RSS.

At $n=10,000$ and $m=12,000$, tiling the phenotype factor reduces median process peak RSS from 1,607.5 to 320.2 MiB across three matched seeds; median time changes from 1.20 to 0.89 seconds. All returned components agree with the dense formula within 2×10^{-12} . Tiled PLINK export

reduces median peak RSS from 811.2 to 308.3 MiB and yields identical BED hashes. Its median time is 0.81 versus 0.88 seconds, with substantial variation across repetitions; the clearest benefit is memory. The dense encoding baseline omits small metadata-writing costs, whereas the tiled writer includes them. These preparation measurements include the resident genotype and interpreter; they are not measurements of an allocation alone.

Table 9 holds $m=12,000$ fixed and extends the matched association comparison to $n=50,000$. It preserves the earlier, pre-optimization implementation with REML-based mixmogam variance fitting. These are single-panel timing and feasibility observations. Their null-chromosome summaries are useful diagnostics, but neither nominal-looking rejection nor greater detection would establish calibration or superiority. Independent reference panels, more phenotype replicates, realistic marker density and stronger donor- sharing audits remain necessary. All four mixmogam KVIK fits at these larger sizes report completed cross-validation and LOCO convergence. Both programs return valid p-values for all markers.

Table 8: HAPNEST-model generation at 12,000 variants: three independent seeds, one thread, warm kernels, with output checksums. Peak RSS includes interpreter and loaded reference. Modes use identical draws for a fixed seed; memory-mapped pages can become resident.

Samples	Backend / output	Median s	Range s	Max MiB
2,000	numba / batches	0.48	0.43–0.81	164.6
2,000	numpy / batches	22.28	21.08–22.28	121.1
10,000	numba / batches	2.31	2.21–2.58	173.5
50,000	numba / array	13.62	12.93–13.94	757.3
50,000	numba / batches	13.32	11.88–13.35	183.0
50,000	numba / memmap	14.24	13.40–14.46	756.6
100,000	numba / batches	26.13	25.73–28.40	198.6

Table 9: Historical pre-optimization HAPNEST-model workloads, $m=12,000$, one realization per cell; mixmogam uses REML. Time is end-to-end seconds; RSS is peak MiB. Inflation and 1% rejection use the chromosome with no generating genetic effects. Structured cells include an environmental effect and two genotype PCs. These descriptive values have no replicate-based uncertainty estimate.

Samples	Cell	KVIK	Seconds	MiB	λ_{GC}	Rejection (%)
10,000	Unstructured	mixmogam	16.08	1316	0.909	1.05
10,000	Unstructured	LDAK	4.20	220	0.931	0.85
10,000	Structure + PCs	mixmogam	18.11	1309	1.086	1.05
10,000	Structure + PCs	LDAK	4.06	221	0.992	1.10
50,000	Unstructured	mixmogam	137.59	2999	0.948	0.90
50,000	Unstructured	LDAK	24.45	627	0.963	1.10
50,000	Structure + PCs	mixmogam	318.02	3234	1.050	1.05
50,000	Structure + PCs	LDAK	35.42	653	0.965	1.30

4.7 Estimator choice and implementation improvements have different effects

The implementation-only comparison held REML, model selection and convergence settings fixed. At 50,000 samples and 12,000 markers, three paired repetitions reduced median complete-fit time from 105.21 to 90.33 seconds without structure, and from 106.29 to 90.31 seconds with structure, environment and PCs: reductions of 14.1% and 15.0%. Across all 12 paired comparisons, including

the 10,000-sample panels, saved variational coefficients and fit diagnostics were exact; the maximum absolute p-value difference was 1.45×10^{-15} . Removing redundant products and allocations helped, but repeated kinship products for variance fitting remained a major cost.

In a separate one-thread comparison on the two 50,000-sample, 12,000-marker panels, choosing HE reduced median complete-fit time from 76.13 to 26.28 seconds and from 73.57 to 27.20 seconds, respectively. These are within-experiment comparisons; the REML times should not be substituted for the baseline of the preceding experiment. HE changed the fitted covariance ratio from 0.49703 to 0.51260 and from 0.50175 to 0.52087. Its time saving is therefore an estimator tradeoff, not a numerically equivalent acceleration of REML.

The independent known-covariance study supports the narrower claim that the projected moment equations target their stated coefficients. Expected unconstrained moments recover v_g and v_e within 1.2×10^{-9} and 1.2×10^{-10} over 12 analytic checks. Across phenotype draws, HE mean ratios at targets 0.3 and 0.6 are 0.3026 and 0.5930 without structure, and 0.2972 and 0.5968 with PCs; corresponding RMSE values range from 0.0508 to 0.0698. At the zero boundary, means are positive (0.0148 and 0.0165), as for the constrained dense comparator. This is finite-sample boundary bias, not evidence of an unbiased heritability estimator. No estimator failures occurred. In the separate global-null scans, both HE and REML reject 0.938% and 0.948% of markers at the 1% threshold in the two cells, respectively, with all 64 fits converged. These phenotype-level summaries are conditional on four fixed genotype panels and cannot establish genome-wide-tail calibration or validity under environmental confounding.

4.8 Twenty thousand markers: speed, memory and numerical stability

All 36 planned fits completed, and every local cross-validation and LOCO fit converged. Table 10 shows the primary resource measurements. Four requested workers reduce mixmogam’s median wall time by factors 1.77 and 1.22 on the unstructured and PC-adjusted confounded panels. At four workers, mixmogam still takes 31% and 7% longer than the pinned LDAK-KVIK Mac executable, respectively, and its median peak RSS is approximately 6.7 and 6.5 times as large. These results support useful optional parallelism on this workload; they do not establish complete efficiency parity.

Table 10: Complete fits at 50,000 samples and 20,000 markers, frozen 2.0.0.dev3 code. Wall time and peak RSS are median [minimum, maximum] of three fresh-process repetitions on each fixed panel. Speedup is relative to the same method’s one-thread median. Timing boundaries are defined in Section 3.4; the swapping limitation below applies to these workstation measurements.

Method	Threads	Wall seconds	Speedup	Peak RSS (GiB)
<i>Unstructured</i>				
mixmogam HE	1	58.84 [43.71, 64.20]	1.00	4.050 [3.113, 4.234]
mixmogam HE	4	33.22 [31.35, 50.30]	1.77	4.369 [3.854, 4.662]
LDAK-KVIK	1	33.49 [33.43, 34.90]	1.00	0.651 [0.651, 0.652]
LDAK-KVIK	4	25.45 [24.51, 28.16]	1.32	0.652 [0.651, 0.652]
<i>Structure, environment, two PCs</i>				
mixmogam HE	1	39.00 [38.69, 51.06]	1.00	4.007 [3.952, 4.414]
mixmogam HE	4	32.06 [27.03, 38.62]	1.22	4.228 [4.086, 5.066]
LDAK-KVIK	1	39.63 [36.71, 40.54]	1.00	0.654 [0.652, 0.656]
LDAK-KVIK	4	30.09 [28.25, 32.61]	1.32	0.653 [0.651, 0.654]

The numerical audit preserves its original elementwise tolerances, $\text{rtol} = 10^{-6}$ and $\text{atol} =$

10^{-8} . All 16 repeated-run comparisons within a method and worker count are exact. All six local four-versus-one comparisons, however, exceed the strict array tolerance; none of the six official comparisons does. Selected exponents and priors, integer iteration counts and convergence flags agree across counts. The largest local absolute differences are 9.7892×10^{-9} for association effects, 3.0826×10^{-11} for standard errors, and 1.4312×10^{-6} for p-values; the maximum absolute log-base-ten p-value difference is 1.15×10^{-5} . All-variant significance masks agree at 0.05, 0.01, 0.001, 0.05/20,000 and 5×10^{-8} in all 30 checks. Unchanged decisions are useful evidence of stability for these particular data and thresholds; they do not turn the six tolerance failures into passes or exclude threshold crossings elsewhere.

Table 11 reports descriptive chromosome-6 diagnostics from four-worker fits. Local HE ratios are approximately 0.52 and 0.50, whereas the official values are approximately 0.25. The different partitioning, large-effect handling and fitted covariance definitions preclude interpreting this discrepancy as bias against a shared target. Both programs detect all ten designated QTLs at both tested stringent thresholds, a saturated result with little power to distinguish methods. Neither nominal-looking bulk rejection nor 3,300 correlated null markers supplies independent replication of a calibration claim.

Table 11: Descriptive diagnostics on the 3,300-marker genetic-null chromosome of each 20,000-marker panel, using four requested workers. Rejections are percentages. Each entry is unchanged across three timing repetitions of one phenotype. Reported h^2 follows each program’s own covariance-fitting convention and is not a common estimand.

Method	Reported h^2	λ_{GC}	$p < 0.05$	$p < 0.01$	$p < 0.001$
<i>Unstructured</i>					
mixmogam HE	0.5224	0.9514	4.879	0.848	0.091
LDAK-KVIK	0.2461	0.9435	4.788	0.879	0.061
<i>Structure, environment, two PCs</i>					
mixmogam HE	0.5049	0.9776	5.576	1.424	0.273
LDAK-KVIK	0.2479	0.9189	4.788	1.333	0.303

The separate cache experiment makes the memory tradeoff explicit (Table 12). Every cached fit retains exactly 4,000,000,000 float32 genotype bytes, while every uncached fit retains zero float-cache bytes. Disabling the cache reduces peak RSS by median paired fractions of 52% and 49%, with 31% and 39% longer wall time. All six cache pairs have exactly equal saved association arrays, variational coefficients and nested fit diagnostics; all eight within-route repeat comparisons are also exact. The 1,000,000,000-byte int8 input, prepared moments, projection coefficients and fitting workspaces remain. Cache-free fitting is therefore a lower-memory option, not an out-of-core analysis. The cache experiment has its own paired baseline and must not replace the primary experiment’s cached timing with a more favourable one.

Resource interpretation has two additional limits. System-wide VM observations spanning the period after primary run nine through the end of all fits record 32,093 additional swap-in pages and 111,568 swap-out pages. They cannot identify which program or fit caused swapping; the wide wall-time and RSS ranges must remain part of the conclusion. Second, the pinned official Mac binary links Accelerate and inspection finds no OpenMP library linkage or matching symbols. Source-level OpenMP directives cannot be assumed active in this executable. These measurements do not establish relative scaling against LDAK’s Linux OpenMP/MKL build.

Table 12: Separate cache experiment at 50,000 samples and 20,000 markers, explicit HE and four workers. Entries are median [minimum, maximum] over three matched repetitions. The only changed fitting option is the standardized-genotype cache budget, 4×10^9 versus zero bytes.

Cache route	Wall seconds	Peak RSS (GiB)
<i>Unstructured</i>		
Cached	30.94 [30.75, 31.38]	4.314 [4.261, 4.639]
Uncached	41.20 [40.59, 41.33]	2.062 [2.061, 2.074]
<i>Structure, environment, two PCs</i>		
Cached	29.88 [29.59, 32.29]	4.085 [4.049, 5.106]
Uncached	41.66 [39.68, 45.02]	2.075 [2.065, 2.083]

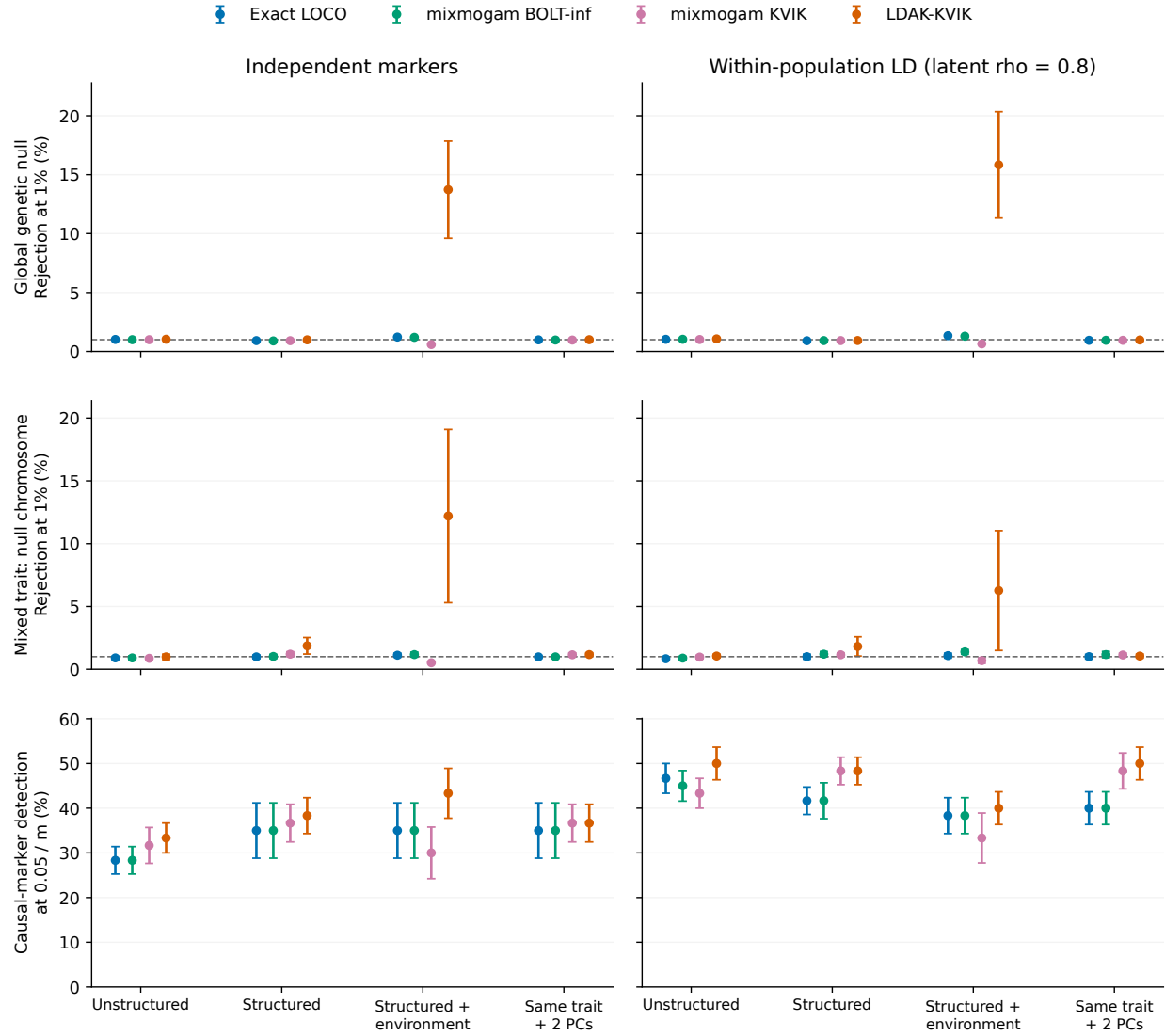


Figure 2: Matched phensim comparison at 800 samples and 6,000 markers. Bars show one Monte Carlo standard error across six independent panels. Dashed lines indicate nominal 1% rejection. The middle row uses only the chromosome without generating effects; the last row measures ten causal markers at $0.05/m$. Omitted environmental effects create associations under the genetic null. Detection in inflated cells is not calibrated power. Columns share scales within each row.

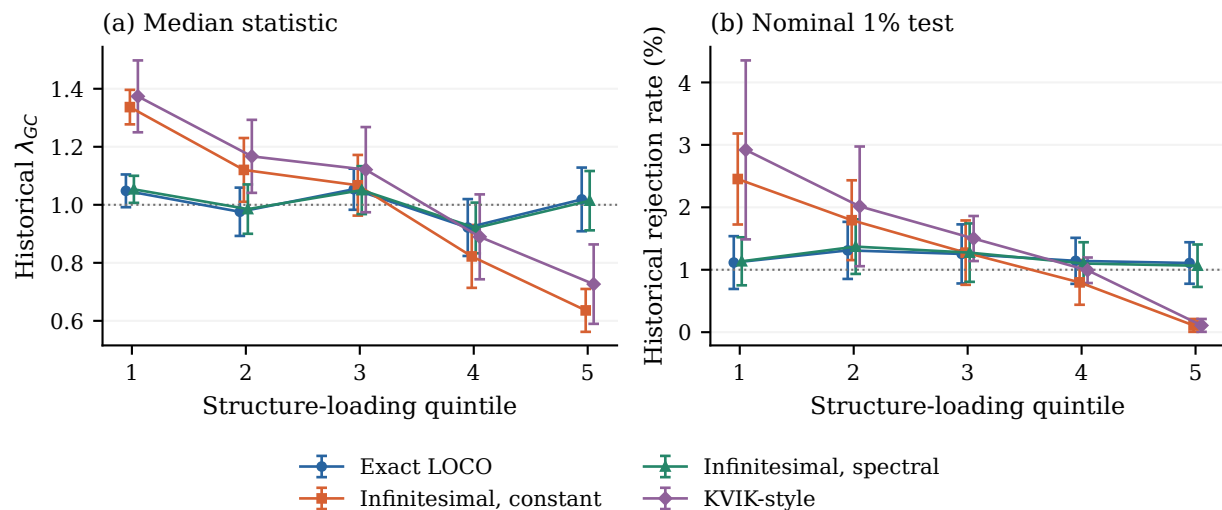


Figure 3: Historical strong-structure diagnostics. Points are means over six phenotype replicates on one fixed genotype panel; bars are 95% Student-*t* intervals for those means. The median statistic and the nominal 1% rejection fraction show why reporting only a genome-wide summary can obscure differences among variant groups. No external program results are included.

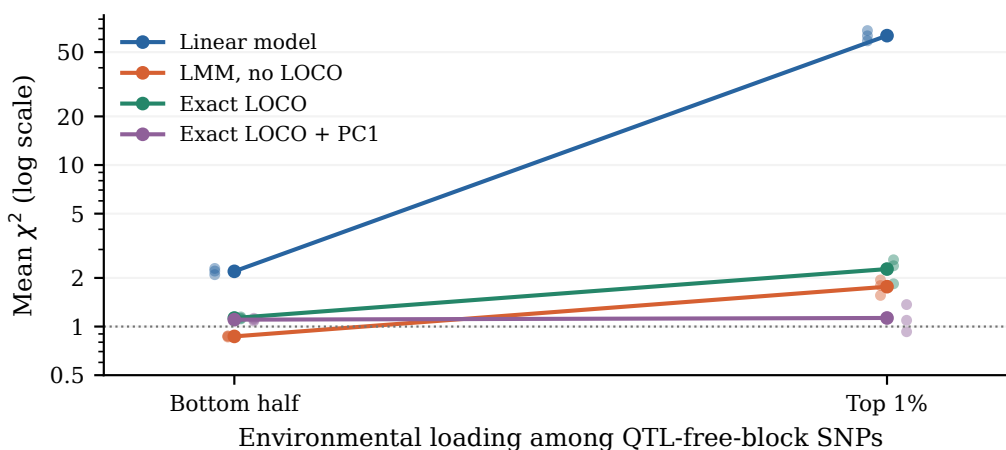


Figure 4: Historical S2 statistics by environmental loading in QTL-free blocks. Large points and lines show means; small points show the three simulation seeds. A logarithmic vertical axis accommodates the large unadjusted mean. PC adjustment reduces the observed gradient, but the polygenic background and small replicate count prevent a strict-null calibration claim.

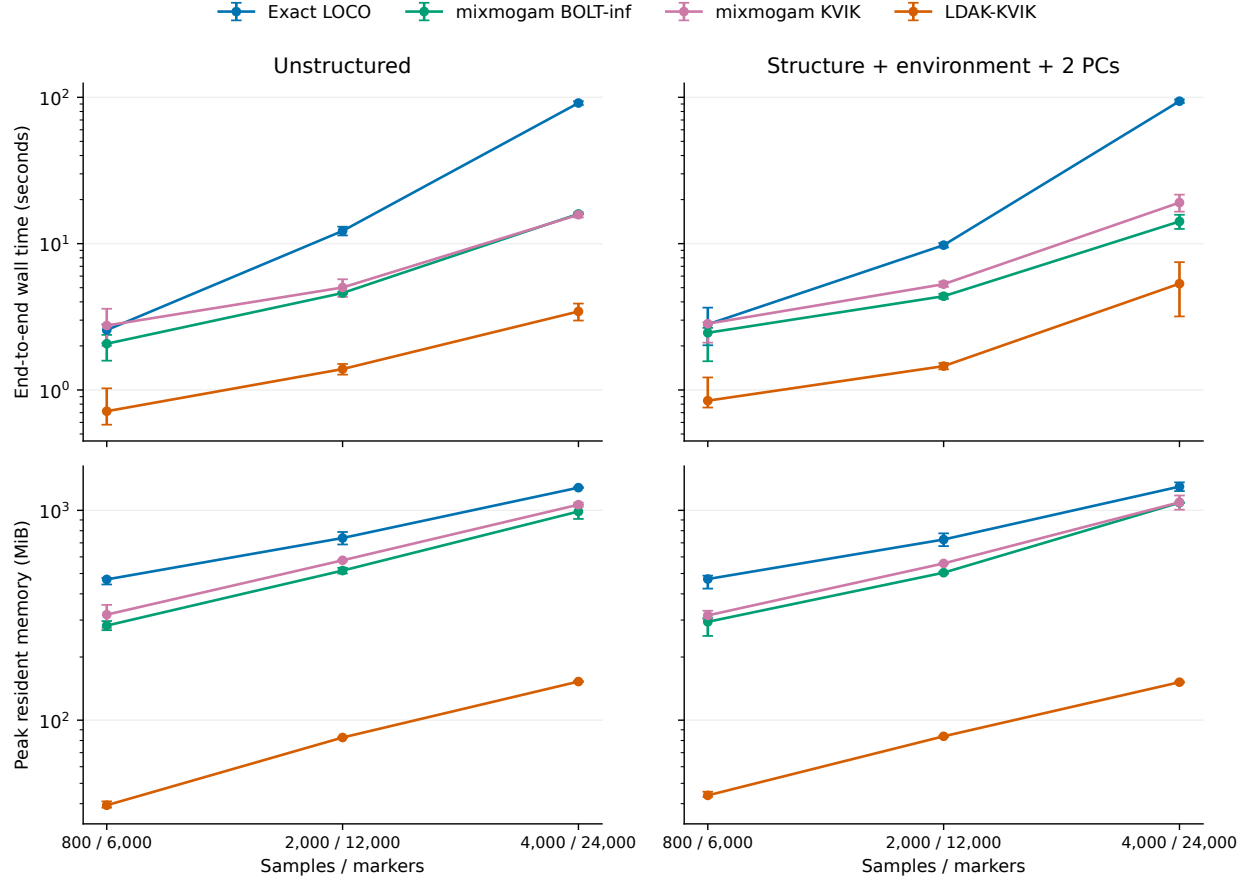


Figure 5: End-to-end time and process peak RSS across three controlled workloads. Points are medians; bars span the observed range across six replicates at $n=800$ and two at $n=2,000$ or $4,000$. Cost axes are logarithmic. Sample and marker counts increase together. These are measurements, not extrapolations or fitted complexity laws.

5 Discussion

5.1 The central distinction is between geometry and inference

Equation (9) explains the observed conditional gradient without invoking a software bug: covariance precision weights variant directions differently. Scalar calibration discards that distinction. The new experiment shows that restoring a few leading directions can substantially reduce the error when population structure is concentrated in those directions and the remaining spectrum is comparatively benign. It also shows why a good average rejection fraction is insufficient.

The result has a deliberately narrow scope. The leading subspace is known exactly, the covariance is fixed, genotypes contain no local LD, and the tested statistic is linear in a Gaussian phenotype before squaring. Estimated covariance, approximate eigenvectors, rare-allele sampling, nonlinear shrinkage, and phenotype-dependent selection all introduce questions not answered by the conditional calculation. The new results justify further evaluation of the spectral correction; they do not justify treating it as a generally calibrated replacement for established GWAS procedures.

5.2 Prediction accuracy and test validity need separate targets

A flexible polygenic predictor can reduce residual variance and thereby improve association power. Yet changing the predictor also changes the null distribution of a test based on its residuals. Calibration inherited from an infinitesimal model may not survive a nonlinear prior, tuning selection, or incomplete optimization. LD Score regression provides a framework for separating components of genome-wide inflation under its assumptions (Bulik-Sullivan et al., 2015); an intercept-based rescaling is not a per-variant guarantee in an arbitrary structured panel.

The same separation applies to reported prediction performance. An internally tuned or in-sample BLUP is not a prospective prediction validation. Any future prediction claim needs a held-out sample design that respects family relationships, ancestry, and the intended target population. A model can improve prediction while leaving a poorly calibrated association statistic, and a calibrated test can sacrifice power through an unsuitable background model.

5.3 The historical evidence changes the questions, not the standard of proof

The environmental experiment illustrates the noncentral mean term in Equation (12). An exposure generated from deme membership is causally external to the candidate variants in the simulation, but it is not statistically independent of them. Calling it independent would remove the very confounding mechanism being studied. A PC defined from the tested genotypes is also a legitimate mathematical stress axis, but answers a different question from a separately generated exposure. Future studies should include both with clearly stated causal and statistical interpretations.

Similarly, a locus label is a scientific choice. A block without a large QTL can tag a QTL across its boundary or contain its own small generating effects. Counting all such discoveries as false positives confounds localization, power to detect diffuse signal, and test calibration. The report therefore keeps strict-null experiments separate from QTL-free-block diagnostics. Real-data significance counts cannot resolve this ambiguity without additional validation.

5.4 Limitations and intended use

The current contribution is strongest as a transparent research implementation and a framework for controlled comparisons. The conditional study contains three small fixed panels, while the broader

archived simulations have three or six replicates and predate important fixes. The new matched-input external comparison has six independent panels per cell at $n = 800$, uses controlled AR(1) rather than demographic LD, and assumes common genetic effects across three discrete populations. It does not establish equivalence of the mixmogam and official LDAK-KVIK models. There is no new validation in human multi-ancestry cohorts, no demonstrated biobank-scale resource profile, and no validation for imputed dosages or strongly imbalanced binary traits. The latter require attention to the likelihood and tail approximation, as illustrated by SAIGE’s logistic mixed-model and saddlepoint approach (Zhou et al., 2018).

The package exposes useful building blocks, but its multiple modes should not be presented as interchangeable. Dense plug-in tests, iterative infinitesimal tests, and nonlinear two-step heuristics differ in what is approximated. Reporting the mode, grouping, covariance estimation, convergence diagnostics, and genotype definition is necessary to make a result interpretable. The most productive next step is to reduce the gap between those explicit contracts and independently validated end-to-end behaviour.

The computational extension narrows a practical gap without removing these inferential qualifications. Reusing work, choosing a cheaper variance estimator, and changing storage are three distinct interventions. The first can preserve a numerical result, the second can change the fitted model, and the third exchanges time for memory. Their evidence should not be collapsed into one speedup from an early implementation to the fastest observed run. At 20,000 markers, memory pressure and floating-point sensitivity are still observable limitations. Keeping REML and one thread as defaults makes the new choices explicit while broader estimator and platform validation proceeds.

6 Future research: hypotheses and decisive experiments

The following directions extend the controlled comparisons in this report. Their order reflects dependence: reliable input identity and reproducible reference comparisons are prerequisites for interpreting a new method’s power or speed. Each direction has a hypothesis and a result that could force it to be revised.

6.1 Priority 1: broaden the matched-input reference benchmark

Question. After input handling and nuisance definitions are matched, do the local implementations reproduce an independent Gaussian reference, and which differences from external programs remain?

Section 3.2 supplies the first matched phensim comparison with official LDAK-KVIK, including independent hard-call and reader checks. The next benchmark should retain that input contract while adding missing calls, admixed and related samples, demographic LD, larger panels and more independent replicates. Dense known-covariance and fitted-covariance cases should remain separate. Comparisons with reference BOLT-LMM, REGENIE, and Quickdraws should document differences in model and default preprocessing rather than assume they implement the same test.

The HAPNEST-model extension makes large synthetic cohorts feasible. The next scientific validation should use independent empirical references: tune copying parameters on one subset, then audit allele frequencies, LD decay, PCs and excessive donor sharing in another. Expansion cannot recover haplotypes absent from its reference. Admixed samples require a model of admixture time and ancestry tract lengths; donor proportions alone do not supply one. Vary sample size and marker density separately and repeat across independently drawn reference panels.

Variance estimation should be an explicit factor in this design. Cross projected HE and REML with polygenic and sparse architectures, known versus omitted environmental means, and increasing covariate leverage. Separate fixed-exponent experiments, where the generating covariance is known, from adaptive exponent selection. Compare raw component bias, boundary frequency, probe uncertainty, association calibration and calibrated power, rather than treating agreement in a reported h^2 as a sufficient endpoint. LDAK’s partitioned estimator and its optional REML revision need separate reference configurations; they cannot serve as interchangeable oracles for the local single-component model. The zero-boundary bias observed here is a specific starting point for that analysis.

Decision criterion. Numerical discrepancies under an identical Gaussian model must be explained at the effect, standard-error, variance-component, and statistic levels before aggregate concordance is accepted. A flat loading gradient in a reference program where the local approximation shows one would restrict the claim to that local procedure. More discoveries alone would not establish better calibration or power. The invalid archived export must not be reused as the matched input.

6.2 Priority 2: test calibration independently of adaptive choices

Hypothesis. An approximation selected on calibration markers retains its claimed accuracy on independent audit markers and independent genotype panels.

A nested design should choose spectral width and scaling using one set of markers and reserve another for evaluation. For LD-containing data, hold out whole blocks where feasible, because random marker splits can make an audit nearly redundant with training. If phenotype-dependent selection is retained, repeat the entire selection and fitting procedure for every null phenotype. Pre-specified strata should include frequency, missingness, LD score, ancestry loading, kinship leverage, and their most relevant intersections. Strict-null traits should first be drawn from a known covariance; fitted-covariance experiments should then add variance estimation as a separate factor.

Decision criterion. Before running the study, specify acceptable relative rejection error at each threshold and use uncertainty intervals across independent phenotype and genotype replicates. A possible initial criterion is a rejection ratio within 0.9–1.1 at $\alpha = 0.01$ in every adequately powered prespecified stratum; it is a proposed tolerance, not a standard already met here. Genome-wide tails need their own error budget and either sufficiently large independent simulations or a justified rare-event method. Bulk calibration or a 3% training-ratio spread cannot stand in for that evidence. Persistent failures in an audited stratum should trigger a more accurate denominator or an explicit unsupported-case result.

6.3 Priority 3: replace spectral heuristics with computable error bounds

Hypothesis. A hybrid method can retain streaming efficiency while bounding denominator error for difficult variants.

Suppose the omitted eigenvalues are known to lie in $[a, b]$, with $a + \delta > 0$, and let $s_j = \|\mathbf{z}_j\|^2 - \sum_{\ell \leq k} a_{\ell j}^2$. The exact retained contribution $q_{j, \leq k}$ then yields

$$q_{j, \leq k} + \frac{s_j}{b + \delta} \leq q_j \leq q_{j, \leq k} + \frac{s_j}{a + \delta}. \quad (15)$$

This interval can be propagated through Equation (10), or through a fixed score numerator, to determine whether an approximation is adequate at the requested threshold. Variants with wide intervals could receive additional spectral work or an iterative exact-denominator solve. Approximate Ritz values alone do not certify the spectrum: residual bounds and uncertainty in the retained

subspace must be included. Equation (15) states a possible mathematical contract, not an implemented certificate.

Iterative solves offer a complementary bound. For $A = \mathbf{K} + \delta \mathbf{I} \succeq \delta \mathbf{I}$, approximate solution $\hat{\mathbf{w}}$, and residual $\mathbf{r} = \mathbf{y} - A\hat{\mathbf{w}}$,

$$|\mathbf{z}_j^\top (A^{-1}\mathbf{y} - \hat{\mathbf{w}})| \leq \frac{\|\mathbf{z}_j\| \|\mathbf{r}\|}{\delta}. \quad (16)$$

A corresponding solve with right-hand side $\mathbf{z}_j$ bounds denominator error. Such bounds may be loose, especially for small δ , but make the numerical target explicit.

Decision criterion. Compare bounds with dense references on small problems, then record how often fallback solves are required, whole-process peak memory, and end-to-end time. The approach is useful only if the bound is honest and selective enough to save work. Even a perfect numerical certificate would leave estimated-covariance and model-misspecification questions unresolved.

6.4 Priority 4: derive calibration for nonlinear polygenic residuals

Hypothesis. Mixture and elastic-net residuals need a calibration that reflects their own response to the phenotype, rather than a transferred infinitesimal factor.

Start with a sequence of models whose complexity increases: fixed ridge shrinkage, tuned ridge, a fixed mixture prior, and then jointly tuned nonlinear models. For each, compare a conditional analytic or influence-function approximation where available with a parametric bootstrap that repeats fitting and hyperparameter selection. Sample splitting or cross-fitting, combined with candidate-region exclusion, may reduce reuse of information between nuisance fitting and testing, but does not by itself establish a null law. Null simulation should include both correctly specified and deliberately misspecified effect distributions, with convergence tolerances varied independently of prior choice.

Decision criterion. An apparent power gain is retained only if it survives equal-calibration comparisons at matched type-I error and can be separated from incomplete optimization or reuse of the same data for tuning. Failure of the infinitesimal factor after nonlinear fitting would support method-specific calibration, not a universal multiplicative repair. Reference-program comparisons are essential because an error in a local reimplementaion is not a property of the published method.

6.5 Priority 5: separate mean confounding, covariance error, and localization

Hypothesis. The benefit and cost of PC adjustment depend on which part of the data-generating process is misspecified, and apparent null-locus inflation depends on the discovery unit.

A factorial simulation should vary ancestry-associated exposure from zero through modest effects to the S2 stress level; covariance fit from known truth to fitted and misspecified kinships; and fixed covariates from none to measured exposure and prespecified PCs. Demographic scenarios should include recent relatives, admixture, unequal ancestry sample sizes, and LD. Generate some outcomes under a strict null and others under sparse, polygenic, and ancestry-dependent effects. Report effect bias, interval coverage, conditional rejection, calibrated power, and between-ancestry performance. PC count should be chosen without looking at the final calibration outcome, or its selection repeated within validation.

For signal-containing traits, distinguish causal-variant recovery, LD-tagged association, and block-level localization. A QTL-free block should be called a false discovery only under a null definition that actually excludes its background and tagging effects. Fixed boundaries, LD-aware

boundaries, and explicitly independent simulated blocks should be compared on the same underlying genotypes where possible.

Decision criterion. If LOCO is calibrated without PCs at modest exposure levels, the strong S2 result should remain a stress-test finding. If PCs reduce bias but also remove genuine ancestry-dependent signal, that trade-off must be reported instead of optimizing λ_{GC} toward one. A block method is better only if it improves the stated localization objective, not merely because relabelling blocks reduces a count.

6.6 Priority 6: make scale and reproducibility part of method design

Hypothesis. Compact hard-call access and better placement of decoding work can reduce the remaining memory gap without forfeiting the speed gained by reusing matrix products.

The immediate experiment is a repeat of the 20,000-marker comparison on an otherwise idle host with enough RAM to avoid the observed swapping, followed by the pinned Linux OpenMP/MKL reference build. Use the same datasets, balanced order, measured CPU time and complete-fit boundaries. Requested thread counts alone do not identify parallel work. Vary sample size and marker density separately, placing marker counts on both sides of the inclusive 4 GB cache boundary with explicit cache budgets. The jump from retained float blocks to repeated decoding is a software threshold, not evidence of a statistical complexity transition.

A subsequent implementation study should compare retained int8 calls with two-bit storage, reuse prepared per-variant moments and covariate coefficients, and profile decoding together with matrix products. Reducing bytes read need not reduce time if unpacking prevents efficient matrix operations. Missing calls, noncontiguous chromosomes, repeated indices and high covariate dimension need independent precision oracles. The current Mp projection store is modest for two PCs but its cost must be included when p grows. Measure setup, variance fitting, prediction and testing separately, alongside total time and peak RSS; retain cold-start and warm-kernel distinctions.

Numerical acceptance also needs a declared use. The present four-thread path preserves tested decisions while failing a strict array tolerance. Future variants should report both effect/standard-error and changes in small p-values, with targeted phenotypes near decision thresholds. Test high LD, small residual variance and difficult covariance spectra before accepting a wider tolerance. Neither a normwise matrix bound nor unchanged decisions on two panels settles this question.

A useful extension would add a carefully specified continuous-dosage interface with imputation-quality handling and independent decoding fixtures. Binary traits require a separate modelling and calibration programme; simply passing zeros and ones to a Gaussian scan is not the proposed extension. Prospective prediction, if developed, should use held-out families or populations and report transportability separately from association power.

Decision criterion. A scalability claim should identify the validated data size, hardware, resource limits, and failure boundary. A speedup is scientifically useful only if the inferential target and accuracy are preserved. Release artifacts should reproduce a compact set of known-truth calculations and input round trips before any larger performance claim is made. A compact backend should advance only when it improves a prespecified time-memory tradeoff over the measured cached and uncached routes, without hiding estimator changes or numerical failures.

Data, software, and reproducibility

The repository is available at <https://github.com/bvilhjal/mixmogam>. This report describes package version 2.0.0.dev4; the known-covariance experiment uses baseline commit 729c8532. Its self-contained driver is `benchmarks/denominator_geometry.py`, with results, source snapshot, and

manifest in 20261003-manuscript-geometry beneath `benchmarks/results/`. The review archive is 20261003-critical-review. Historical inputs retain their original source snapshots. The excluded external-program archive is also preserved for provenance; its results are not incorporated into the revised displays.

The matched phensim study is archived as 20261003-phensim-kvik, using mixmogam baseline db232216 and phensim baseline 4939124e with the explicitly snapshotted simulator extensions. The generating and analysis driver is `benchmarks/kvik_simulation.py`; the frozen plan, full package sources, complete inputs, raw p-values, commands, logs, diagnostics and resource measurements accompany its results. The separate `report/make_kvik_figures.py` renders this evidence without rerunning any association test.

The workload extensions are listed in the report evidence map. Large BED files remain local, with hashes, phased references and generating sources retained in their archives. HAPNEST resource summaries here link to full phensim source/reference archives. The simulation-only and association measurements must not be combined into a claim about the cost of a whole-genome analysis at 100,000 samples.

The implementation-only comparison is frozen in 20261003-kvik-efficiency. The estimator comparison, 20261003-kvik-he-efficiency, also retains the independent variance and global-null validation.

The 20,000-marker inputs and their all-call audit are archived as

20261003-hapnest-kvik-n50000-m20000.

The thread and cache experiments are in 20261003-kvik-20k. They retain commands, raw measurements, association arrays, model diagnostics and verification records, including numerical failures and VM observations. Their frozen source version is 2.0.0.dev3. This report’s version bump does not relabel those computations.

The table generator, `report/make_efficiency_tables.py`, reads saved individual measurements and never reruns a model. Its `efficiency_table_manifest.json` records input and output hashes.

`report/README.md` gives the reproduction commands, exact archive identities, and evidence map. The figure generator reads recorded results and does not silently rerun old analyses. The figure and report manifests identify source hashes; the bibliography metadata records verified DOI identities. No new human-participant data were collected for this revision. The new experiments are entirely simulated. Historical RegMap results are not used to support current performance claims.

A Useful identities and interpretation checks

A.1 Whitening explains the known-shape $\mathbf{F}$ reference

Let $L = \mathbf{H}^{-1/2}$ and let Q span $L\mathbf{X}$ orthonormally. The whitened residual phenotype is $\mathbf{r} = (\mathbf{I} - QQ^\top)L\mathbf{y}$ and the residual candidate is $\mathbf{v}_j = (\mathbf{I} - QQ^\top)L\mathbf{z}_j$. Under the known-shape Gaussian null, the squared projection onto $\mathbf{v}_j$ and the remaining residual sum of squares are independent scaled chi-squares with one and $n - p - 1$ degrees of freedom. Their ratio is Equation (7). When L is estimated from $\mathbf{y}$, this fixed-linear- transformation argument no longer proves the reference law. It explains precisely where the plug-in approximation enters.

A.2 Residual coordinates for permutation

Choose an $n \times (n - p)$ orthonormal matrix B spanning the complement of $L\mathbf{X}$. With known Gaussian covariance shape,

$$\mathbf{t} = B^\top L\mathbf{y} \sim N(0, \sigma_g^2 \mathbf{I}_{n-p}), \quad \text{Var}\{(\mathbf{I} - QQ^\top)L\mathbf{y}\} = \sigma_g^2(\mathbf{I} - QQ^\top). \quad (17)$$

The entries of $\mathbf{t}$ are exchangeable; the n entries of the projected residual generally are not. A valid residual-coordinate transformation therefore permutes $\mathbf{t}$ and maps it back through B , rather than permuting arbitrary sample-coordinate residuals. The implementation uses orthogonal transformations without storing an additional full complement matrix in the production path. Estimated covariance, non-Gaussian errors, and repeated model selection require separate investigation; algebraic exchangeability under a known model should not be advertised as an unconditional guarantee.

A.3 BLUP scale and a marginal Bayes factor

The genetic conditional mean uses $\text{Cov}(\mathbf{u}, \mathbf{y}) = \sigma_g^2 \mathbf{K}$ and $\text{Var}(\mathbf{y}) = \sigma_g^2 \mathbf{H}$, hence

$$\hat{\mathbf{u}} = \sigma_g^2 \mathbf{K}(\sigma_g^2 \mathbf{H})^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{K}\mathbf{H}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}). \quad (18)$$

This identity is conditional on fitted parameters when used in the package. In-sample fitted genetic values do not measure prospective prediction accuracy.

For a marginal effect estimate $\hat{\alpha}$ with sampling variance v , assume $\hat{\alpha} \mid \alpha \sim N(\alpha, v)$ and an alternative prior $\alpha \sim N(0, W)$. Against a point null at zero, the approximate Bayes factor is

$$\text{BF}_{10} = \sqrt{\frac{v}{v + W}} \exp \left\{ \frac{\hat{\alpha}^2 W}{2v(v + W)} \right\}. \quad (19)$$

The prior variance W has the units of the squared effect and must be chosen explicitly. Posterior odds additionally require prior odds; normalizing marginal factors across correlated variants does not by itself yield a multi-variant fine-mapping posterior. The corrected API therefore no longer presents an RSS-based pseudo-score as a posterior probability.

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